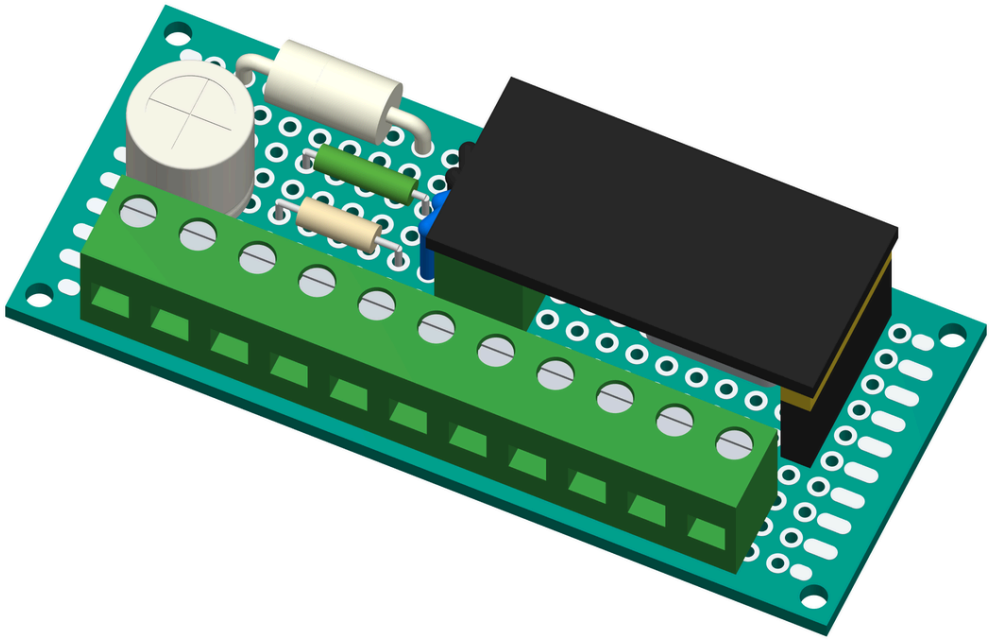
The boards

How they work, how to build them.



Designed by Athanasia Nikolova

All data come from tests and many real builds, not only from theory. v1.2 · November 2026

Teacher only



One thing per page

One action, one big picture. Short sentences, much white space.



Color is never the only clue

Every color also has a number, a letter or a name next to it. The part colors stay distinct with color-vision deficiency (Okabe-Ito palette).



Clear print

Instructions at 14 pt, everything else at least 12 pt, in Atkinson Hyperlegible. The text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).



For a screen reader

The PDF is tagged (PDF/UA-1): headings, lists and reading order are marked, and every picture has a text alternative that is read aloud.



Print it right

In color, at 100 % - not 'fit to page'.



Pace

One page, then the action, then the next page. For a student who tires quickly: one lesson per session. The check at the end of each lesson is a good place to stop.

Safety first



Grown-up nearby

Work with your teacher or mentor in the room.



Hot!

The iron is hot. Put it back in its stand every time.



Fresh air

Solder smoke is not good to breathe. Open a window.



Eyes

Wear glasses when you cut wire legs - they fly.

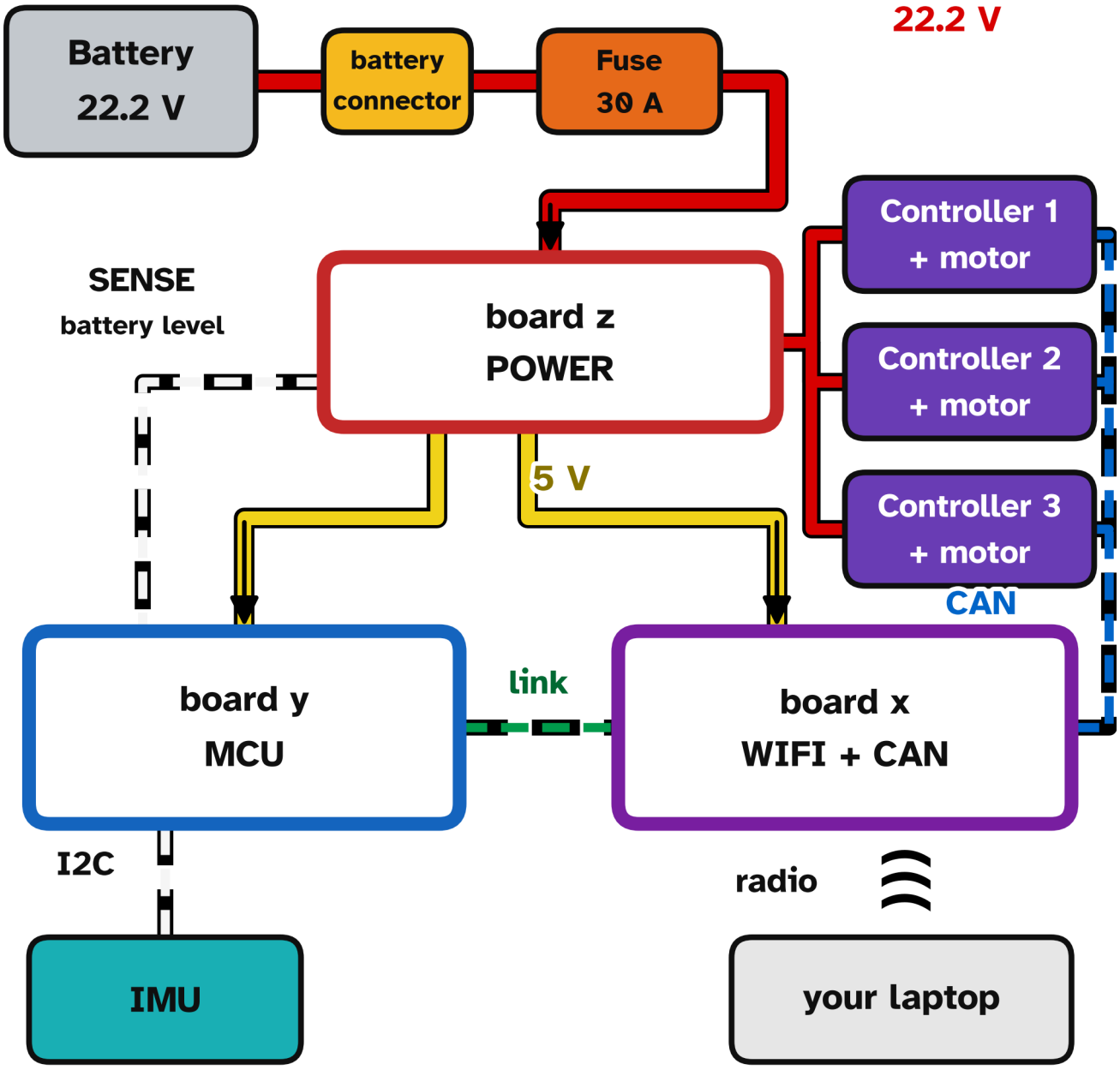


Power last

Connect the battery only after the check pages are done.

The big picture

Thick lines carry power. Dashed lines carry messages.



Power: 22.2 V and 5 V



Battery

6 cells x 3.7 V = 22.2 V (25.2 V when full).



Fuse 30 A

If too much current flows, it melts and cuts the power.



board z

Gives 22.2 V to the 3 motor controllers.



DC-DC

Turns 22.2 V into 5 V for boards y and x.



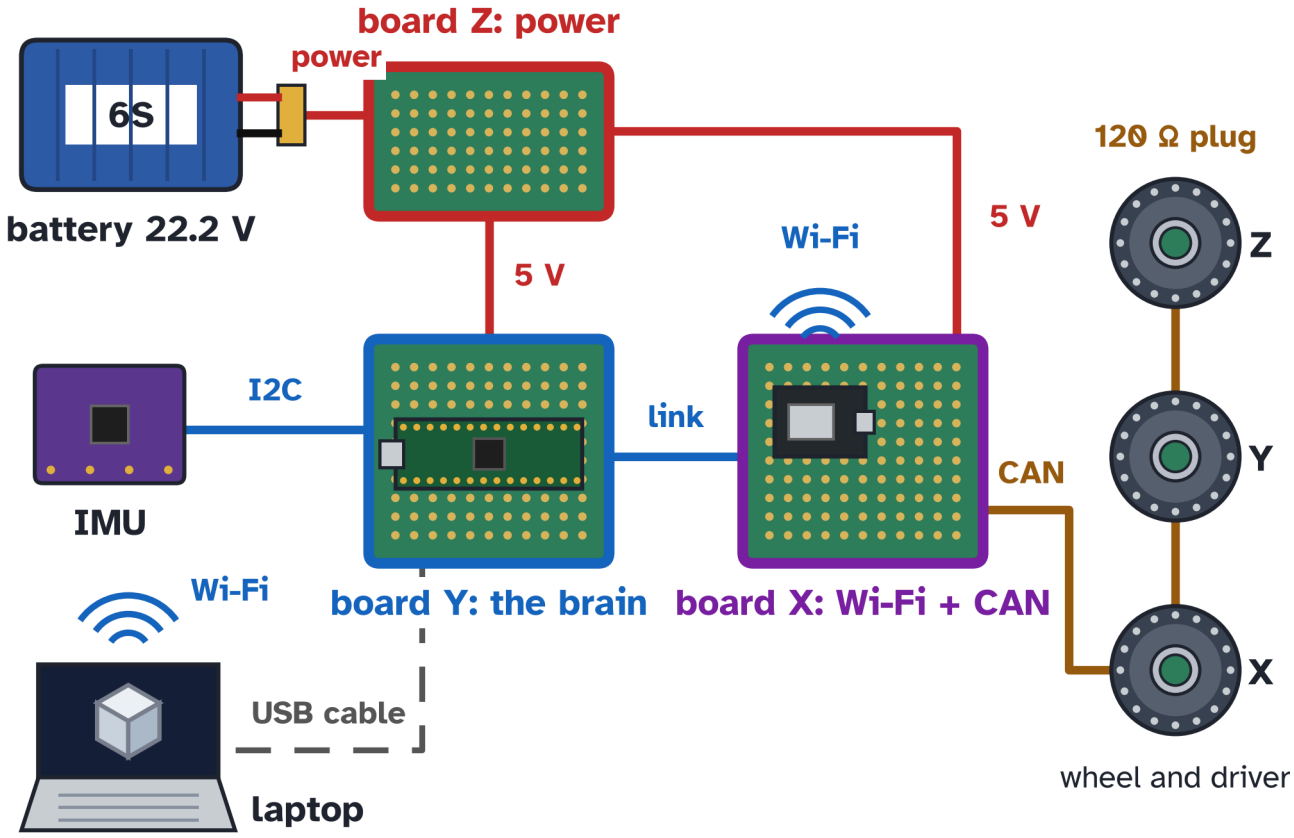
MCU

Makes its own 3.3 V from the 5 V.

How much current?

Balancing still: about 0.9 A. One wheel at full power: 10 A. All three at the worst moment: 30 A - that is why the fuse is 30 A.

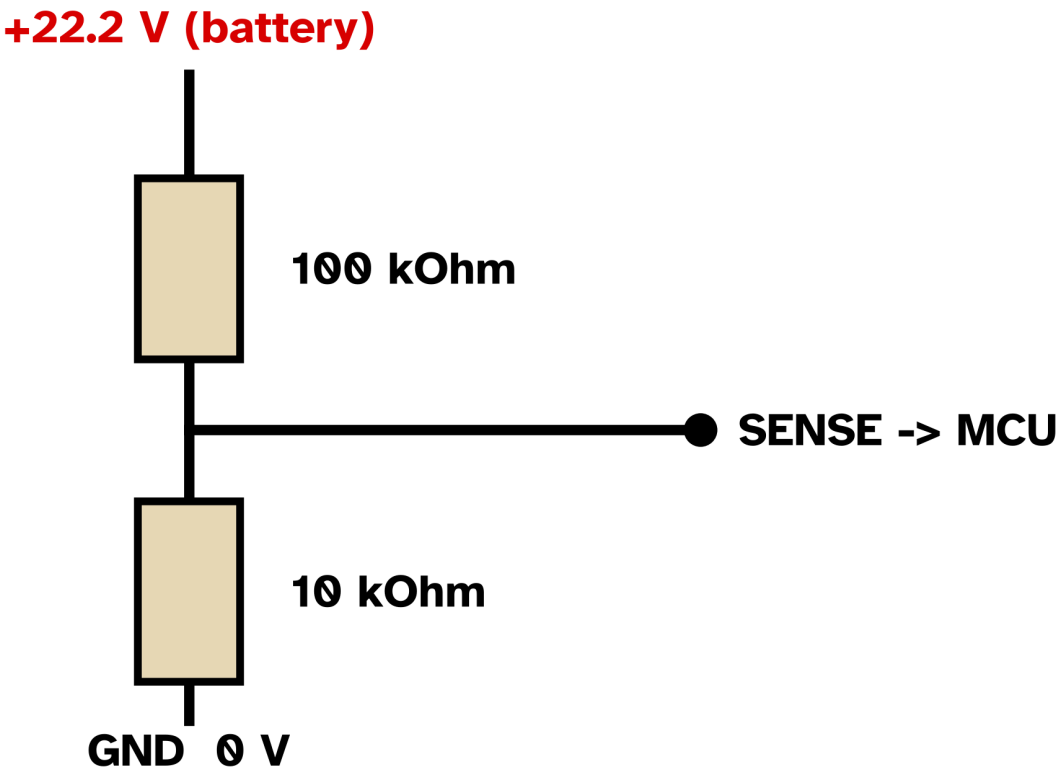
Messages: who talks to whom



- **MCU**: the brain. It decides everything.
- **IMU to MCU**: how the cube is tilted (I2C wires).
- **SENSE to MCU**: how full the battery is.
- **MCU to CAN**: tells each motor controller how hard to turn.
- **MCU and Wi-Fi board**: the link cable between boards y and x.
- **Wi-Fi board and laptop**: by radio - you see the data, you send commands.

How the MCU reads the battery

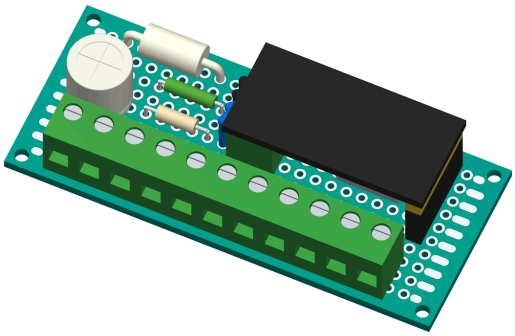
The MCU can read at most 3.3 V. Two resistors make the battery voltage 11 times smaller.



battery	on SENSE	
25.2 V x 10 / 110	2.29 V	full
22.2 V x 10 / 110	2.02 V	normal
19.8 V x 10 / 110	1.80 V	empty - stop!

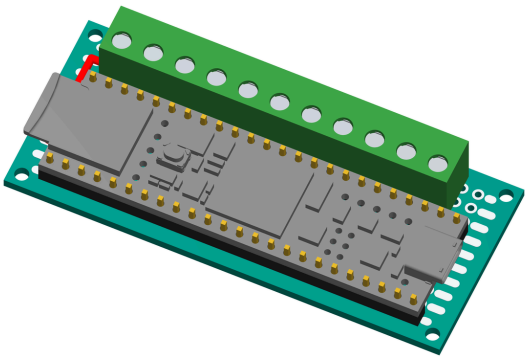
Meet the three boards

Each board has its own color on every page, and its letter.



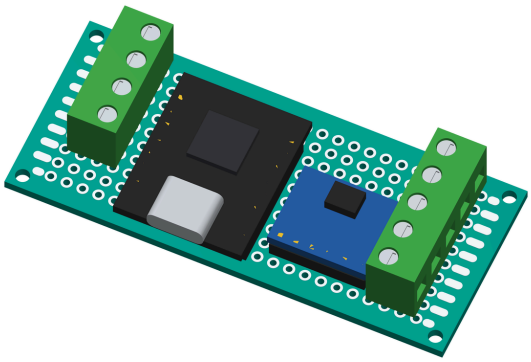
board z · POWER

Takes the battery (22.2 V), feeds the 3 motor controllers and makes 5 V for the other two boards.



board y · MCU

The brain: the MCU board reads the IMU and the battery voltage and decides how the wheels must turn.

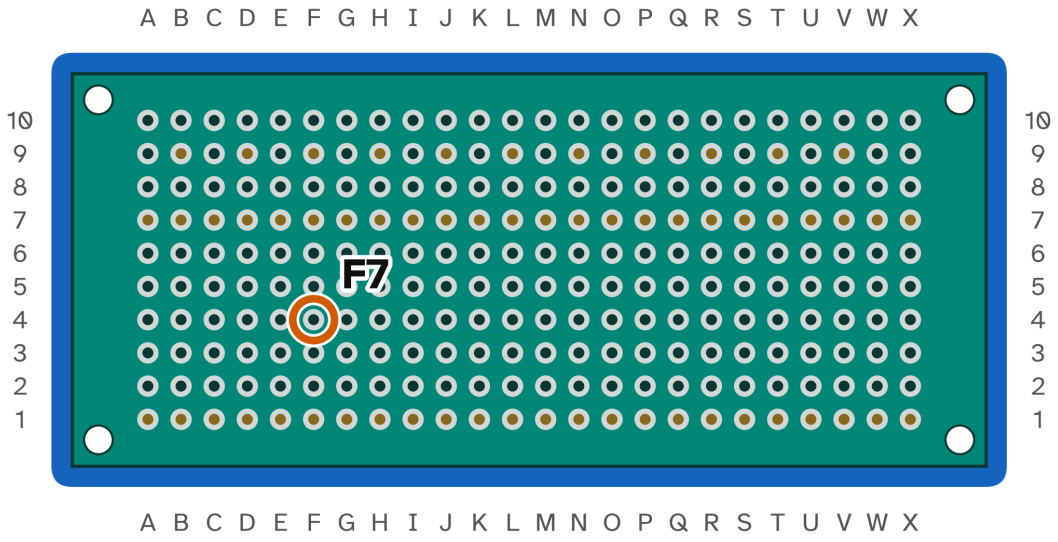


board x · WIRELESS + CAN

Talks to your laptop by radio (Wi-Fi board) and to the 3 motor controllers over the CAN bus (CAN board).

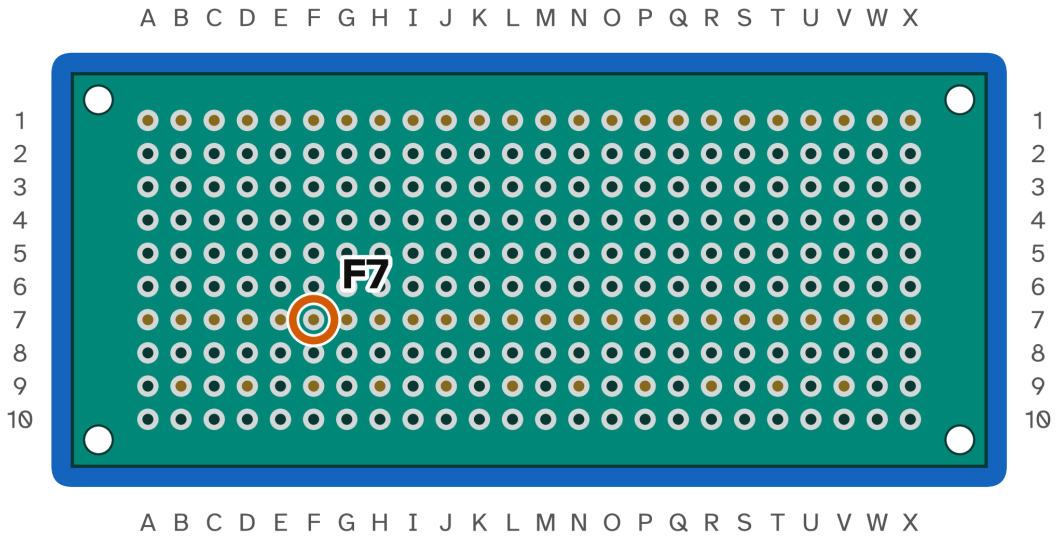
Find a hole

Letter = row (A … X). Number = column (1 … 10).



Hole F7: row F, column 7.

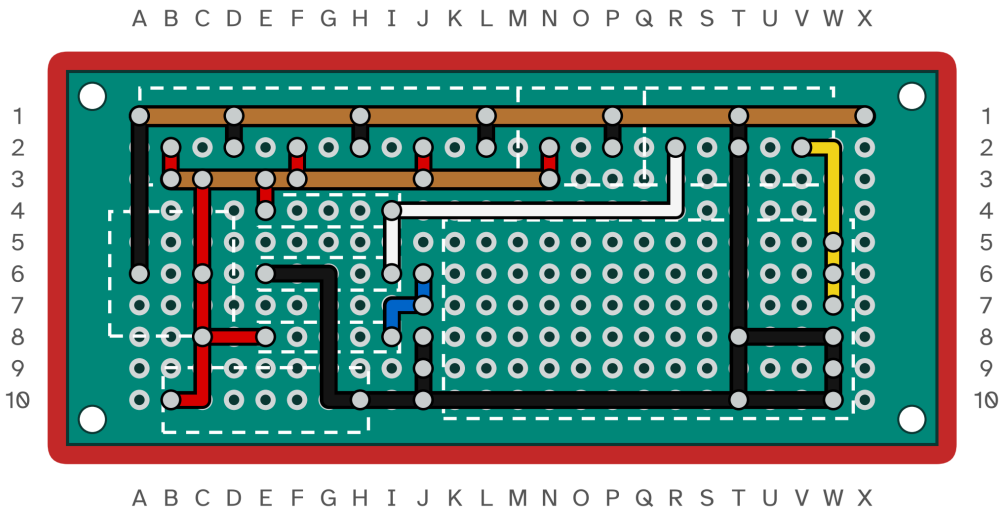
Turned over, the letters stay in place. The numbers go the other way:



Wire colors

Every wire is BARE tinned copper. The colors are only in the drawings - so they are easy to read.

- **red**: “+22V battery power”
- **black**: “GND ground, 0 V”
- **yellow**: “+5V”
- **white**: “message”
- **blue**: “message”
- **green**: “message”
- **copper**: bus bar (board z), soldered all along



- Each wire has a number, outside the board, with a thin line to it.
- A dot on a wire = a solder joint: its two ends, and every pin it is soldered to on the way.

The order of work

The same on every board.


1 parts


2 solder


3 power wires


4 signal wires

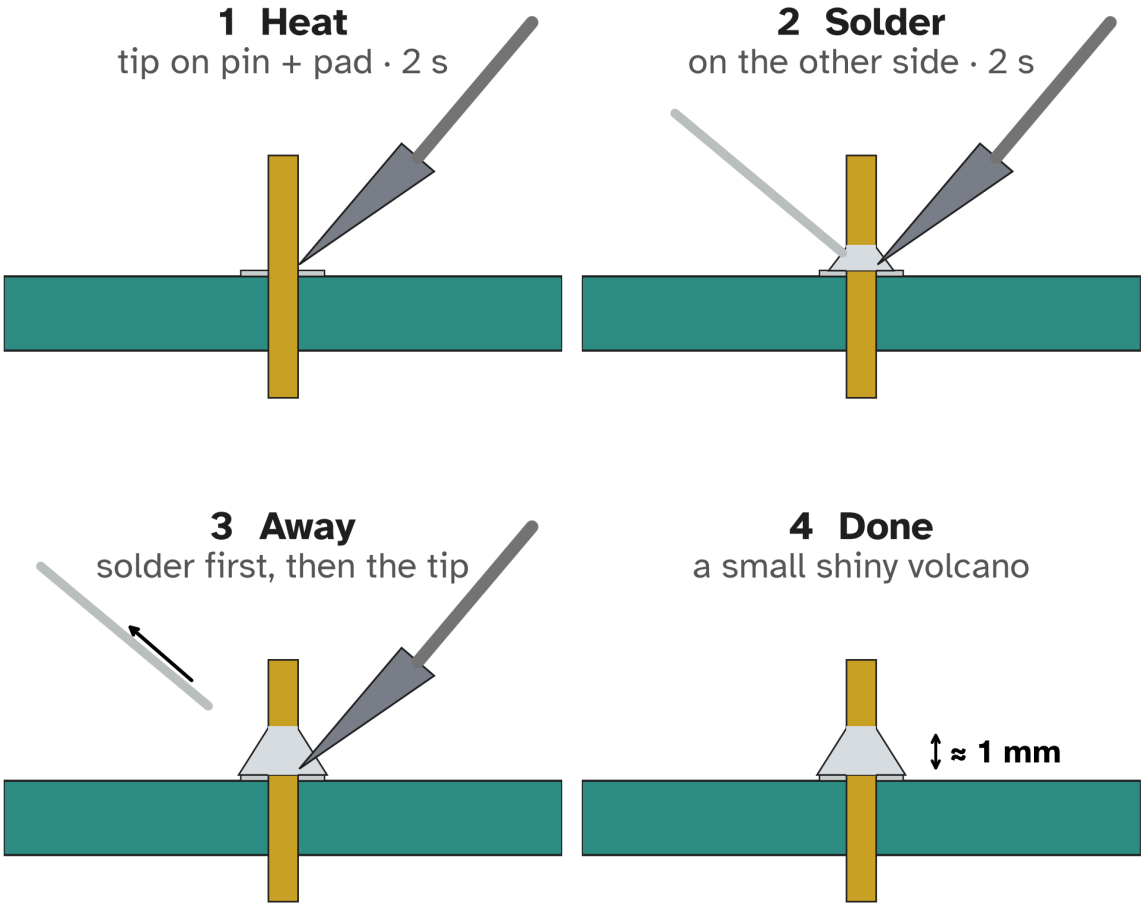

5 check

1. Parts: put each part in its holes, on the top side.
2. Solder: turn the board over. Solder every pin.
3. Power wires: bus bars, red, black, yellow - first, they lie lowest.
4. Signal wires: white, blue, green - after the power.
5. Check: multimeter - beep where it must, silence where it must not.

All wires go on the BOTTOM side (the copper side). All of them are bare wire: they never touch each other, nor a pin of another wire.

Soldering school

Every joint is made the same way: heat, solder, away.



- Solder melts only on hot metal. Heat the metal, not the solder.
- A good joint is a small shiny volcano round the pin.
- Practice 10 joints on a spare board before the real one.
- The iron at 350 °C. Wire takes the heat away: hold the tip a little longer, never more than 3-4 s on one joint.

Materials for good joints



iron · solder, wire · cutter · magnifier · glasses · fan

- Solder: tin-lead 60/40 or 63/37, 0.6-0.8 mm, with a flux core. Wash your hands after.
- Optional - flux: no-clean flux gel in a syringe. A tiny drop on the pad and on the wire before each joint: the solder flows better and the joint shines.
- Optional - contact cleaner: isopropyl alcohol (IPA, 99 %) in a dropper bottle. Clean the pads and the old copper before you solder; clean the flux off at the end.
- Wire: bare tinned copper, solid, 0.6 mm (22 AWG).
- Iron tips: a fine conical one for the pins, a 2 mm chisel one for the wire.
- Brass wool to clean the tip; tip tinner when the tip turns black.
- Desoldering pump and wick, for mistakes.
- Flat-nose pliers, side cutter, tweezers.
- Helping hands with a magnifier, or a board holder - and a lamp.
- An old toothbrush, with the alcohol, to scrub the board clean at the end.
- Glasses and a small fan: the smoke goes away from your face.

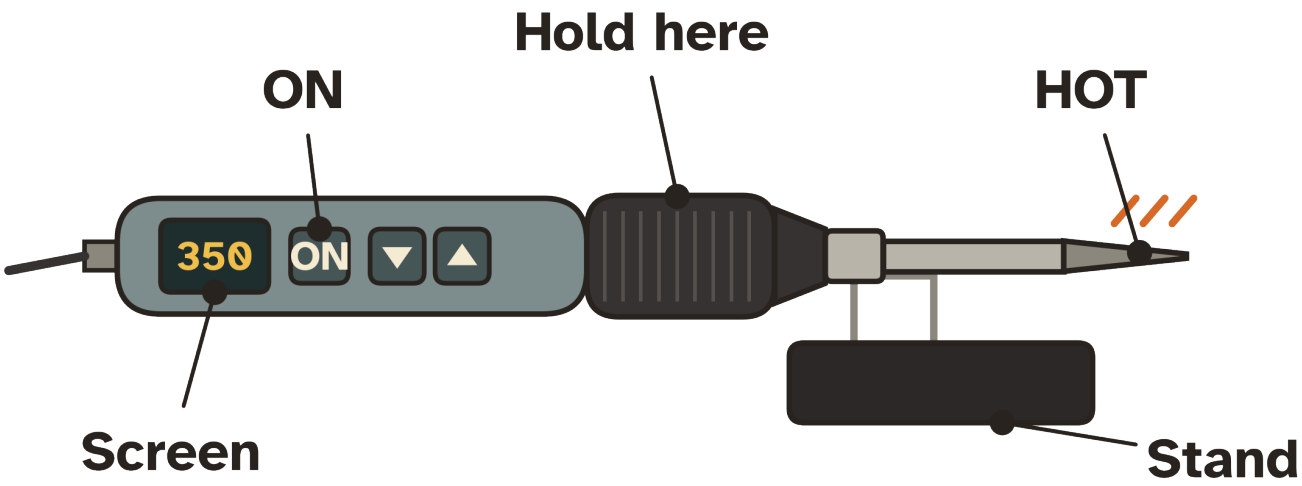
Bare wire, step by step



1. Straighten it: hold both ends with pliers and pull.
2. Tin it: a little flux, then run the hot tip with solder along it.
3. Bend every corner with the flat-nose pliers, right over the center of a hole.
4. Lay it flat on the board. Solder both ends first, then every dot on the way.
5. Cut the spare end short, the flat side of the cutter toward the joint.
6. Check: it touches no other wire and no pin of another wire.

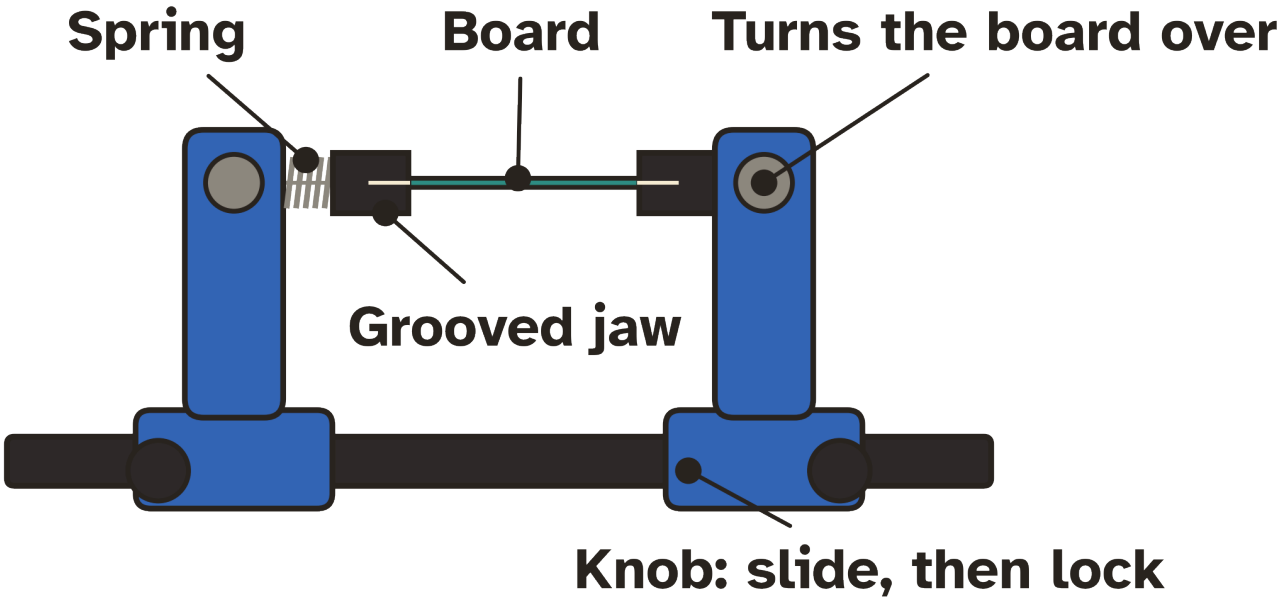
Your tools: the soldering iron

Set the tools out before you start.



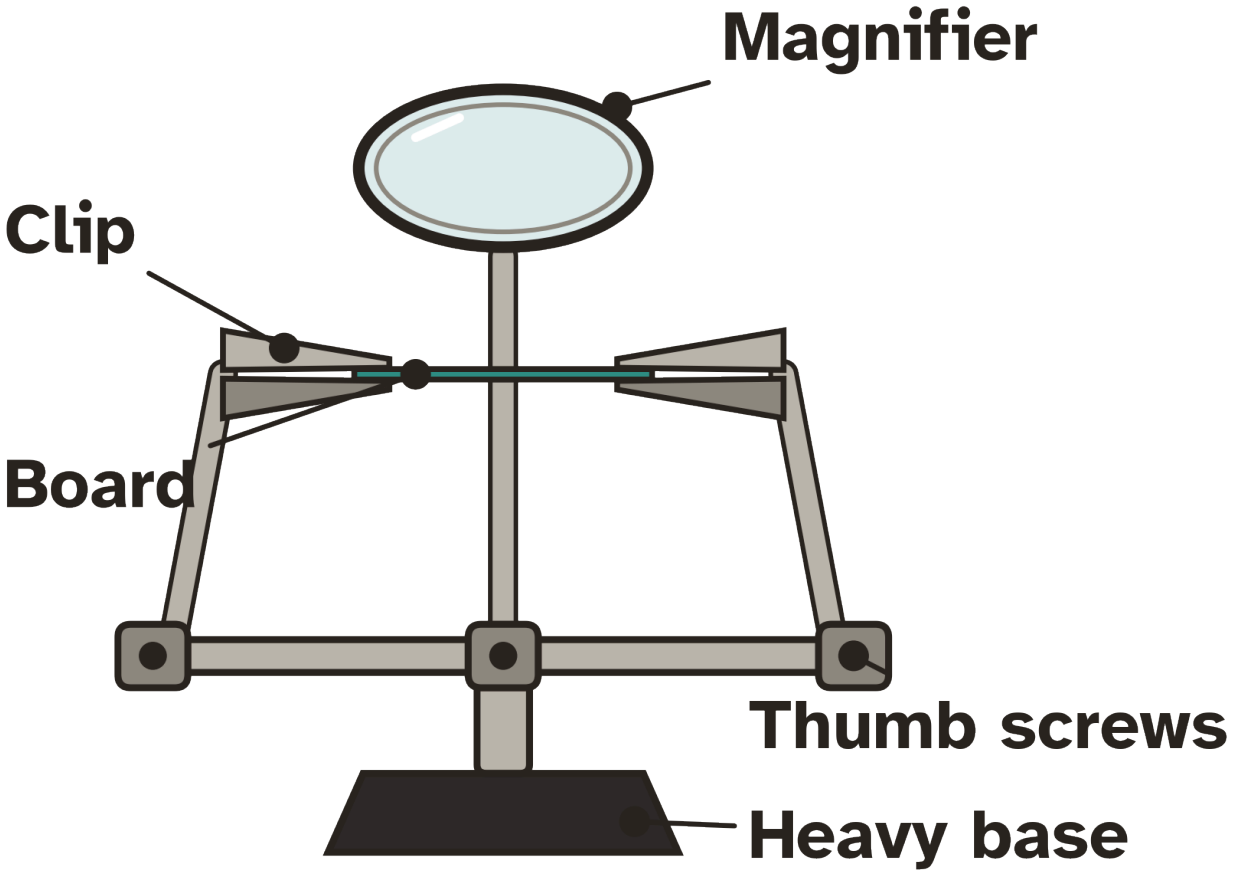
- Conical tip for the pins, 2 mm chisel tip for the wire. Set it to 350 °C.
- Back in the stand after every joint.

Your tools: the board holder



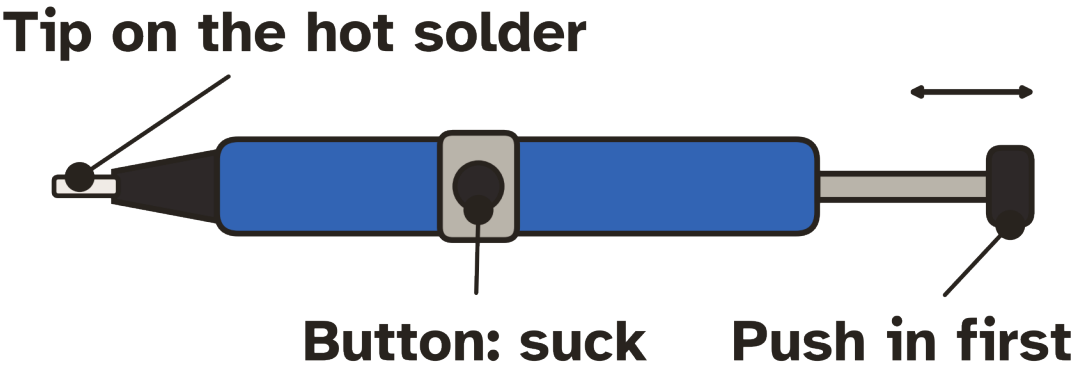
- The board sits in the two grooved jaws. Turn it over without taking it out.

Your tools: the helping hands



- Two clips hold the board. Look through the magnifier at every joint.

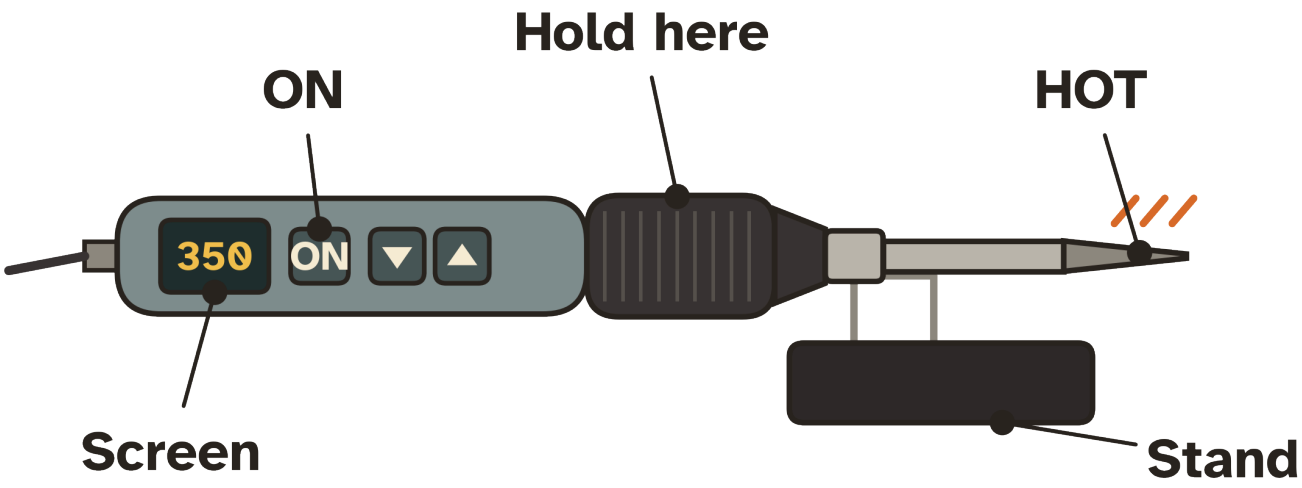
Your tools: the desoldering pump



- For mistakes: push the knob in, heat the solder, tip on it, press the button.

Hold the iron

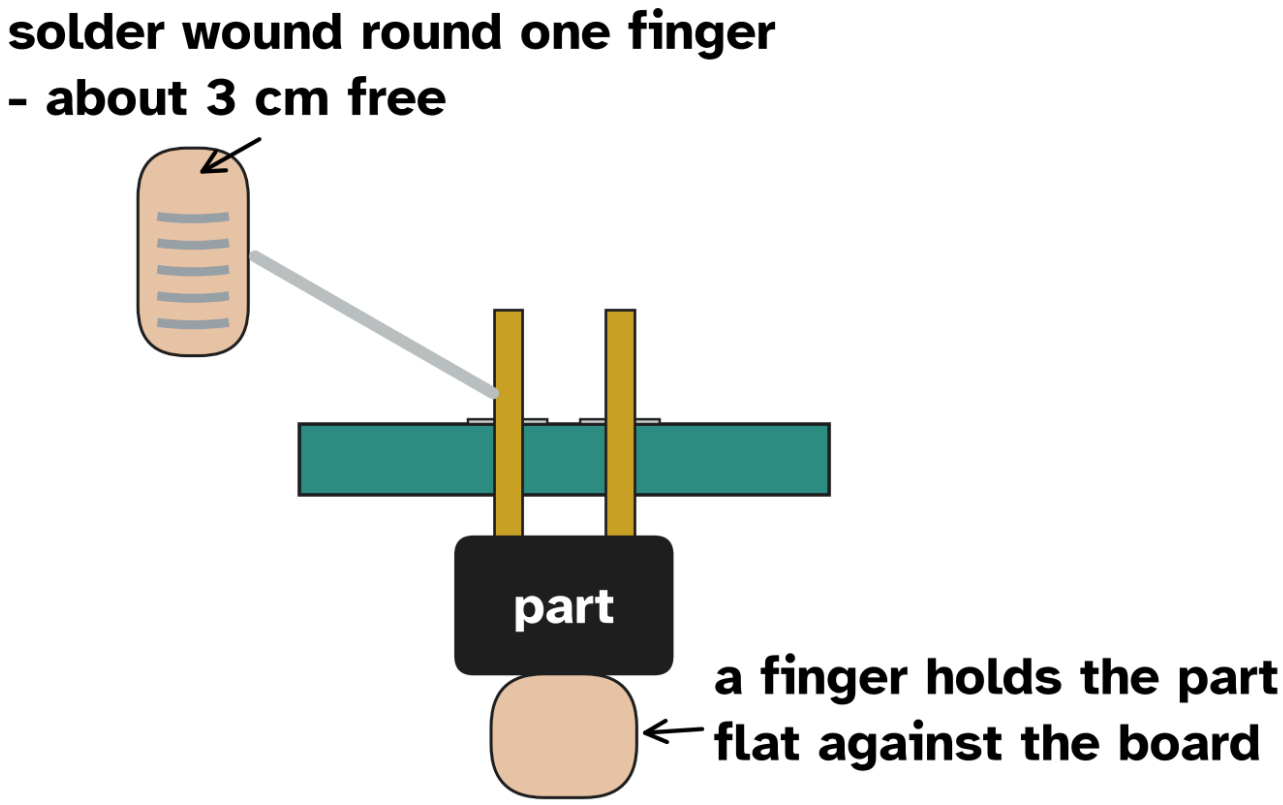
Like a pen - on the grip, never on the metal.



- Hold it on the soft grip, like a pen.
- Rest your hand on the table: the tip stays steady.
- Back in the stand after every joint.
- Clean the tip in the brass wool. A clean tip is shiny.

Hold the solder

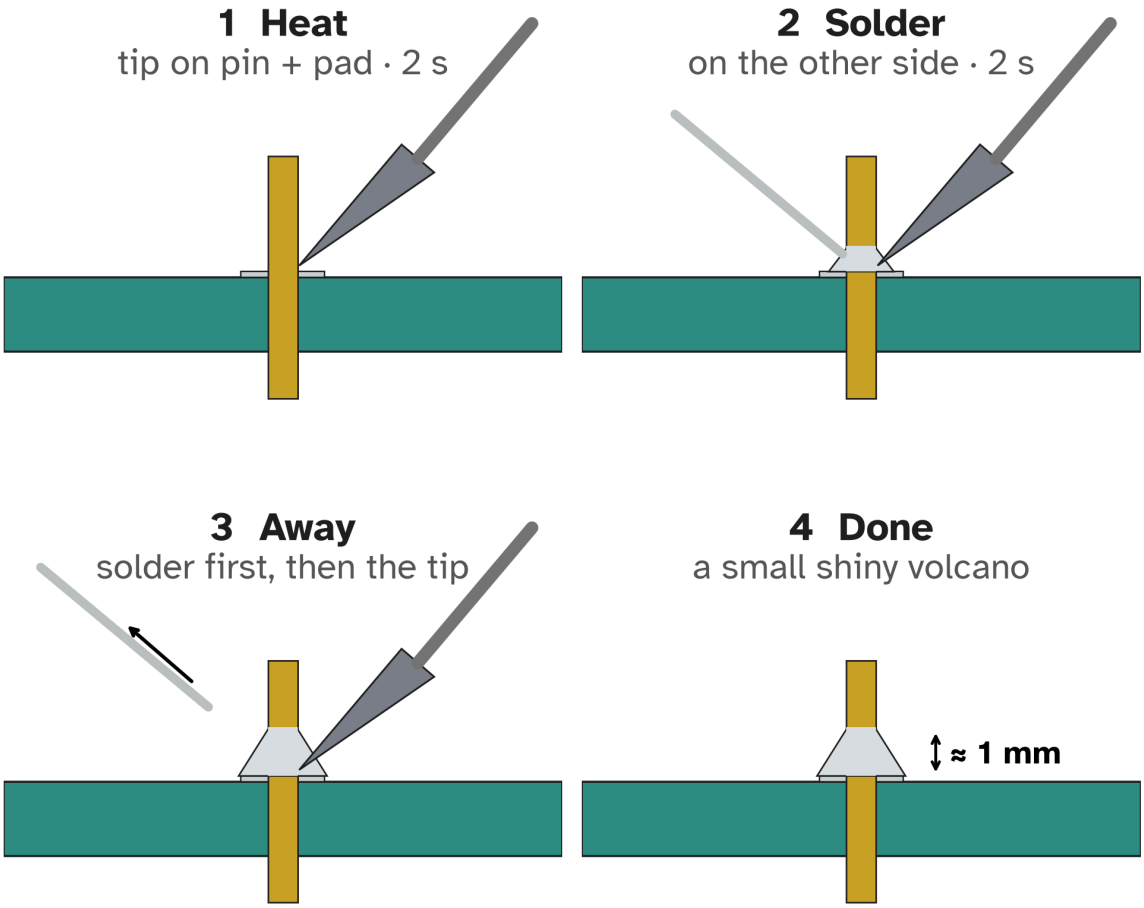
One hand holds the iron. The other hand does two jobs.



- Wind the solder round one finger. Leave about 3 cm free.
- With another finger of the same hand, press the part flat against the board.
- Feed the solder from that hand, a little at a time.

One joint, step by step

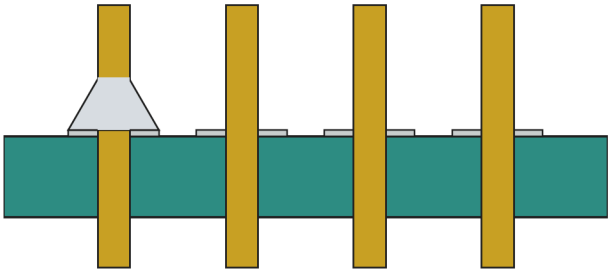
1. Tip on the pin AND the pad together. Count 2 s.
2. Touch the solder on the other side of the pin. Count 2 s more. Add as much as it needs: the pad covered and about 1 mm up the pin.
3. Take the solder away first, then the tip.
4. Do not move the part for a few seconds: the solder is setting.



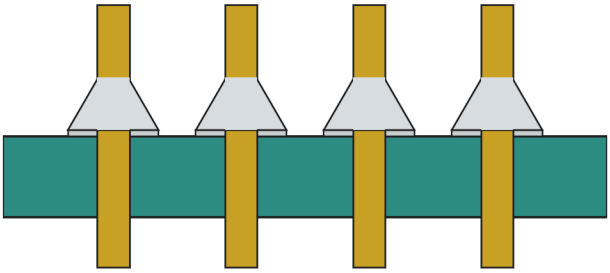
First one pin, then the holder

1. Solder ONE pin only, while your finger holds the part flat.
2. Look from the side: is the part flat on the board? If not, heat that pin and press again.
3. Now put the board in the holder (helping hands) and turn on the lamp.
4. Solder every other pin, one at a time, carefully.

1 One pin only
a finger holds the part flat



2 In the holder: all the others
one at a time, with a lamp

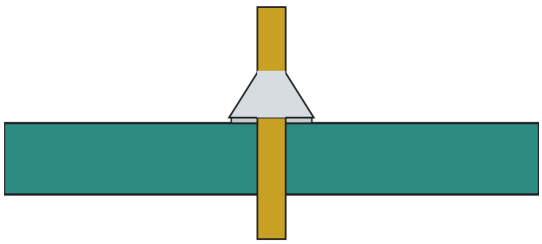


Good or not?

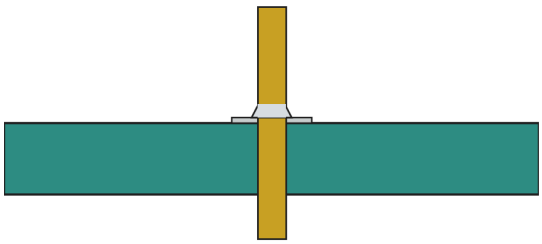
Look at every joint with the lamp.

- Too little: heat again and add a little solder.
- Too much: heat it and pull the extra away with the clean tip.
- Bridge: heat it and pull the tip between the pins, or use the pump or the wick.
- Dull or grainy: heat again for 2 s.

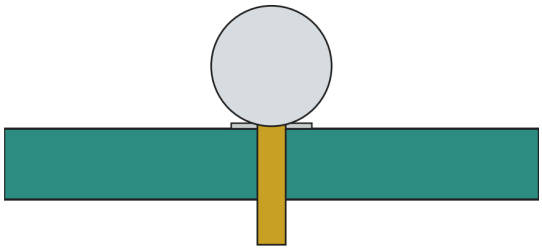
Good
pad covered, ~1 mm up the pin



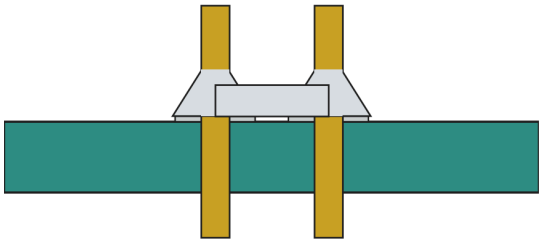
Too little
heat again, a little more



Too much
a ball - pull the extra off



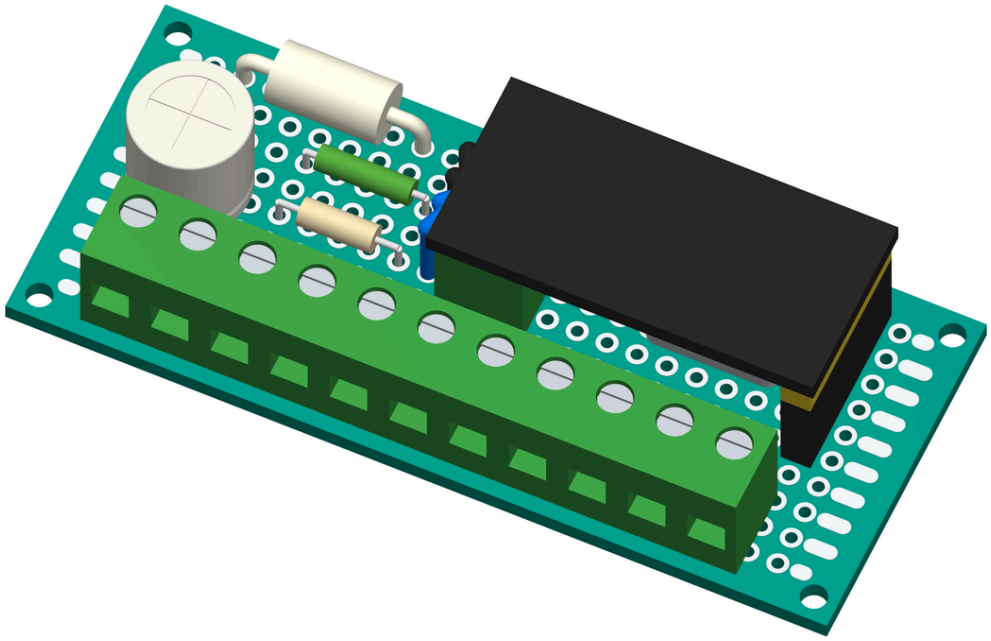
Bridge
two pins joined - never!



board z · POWER

Takes the battery (22.2 V), feeds the 3 motor controllers and makes 5 V for the other two boards.

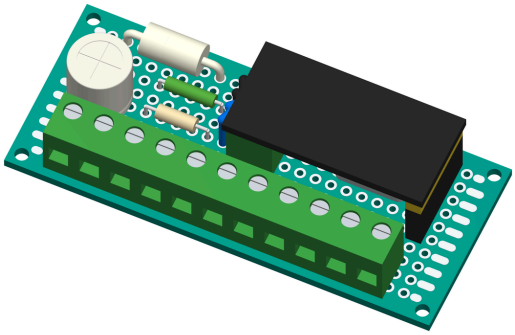
board z



10 parts · 18 wires · 2 bus bars

Face z of the box. Find the letter Z on the box wall.

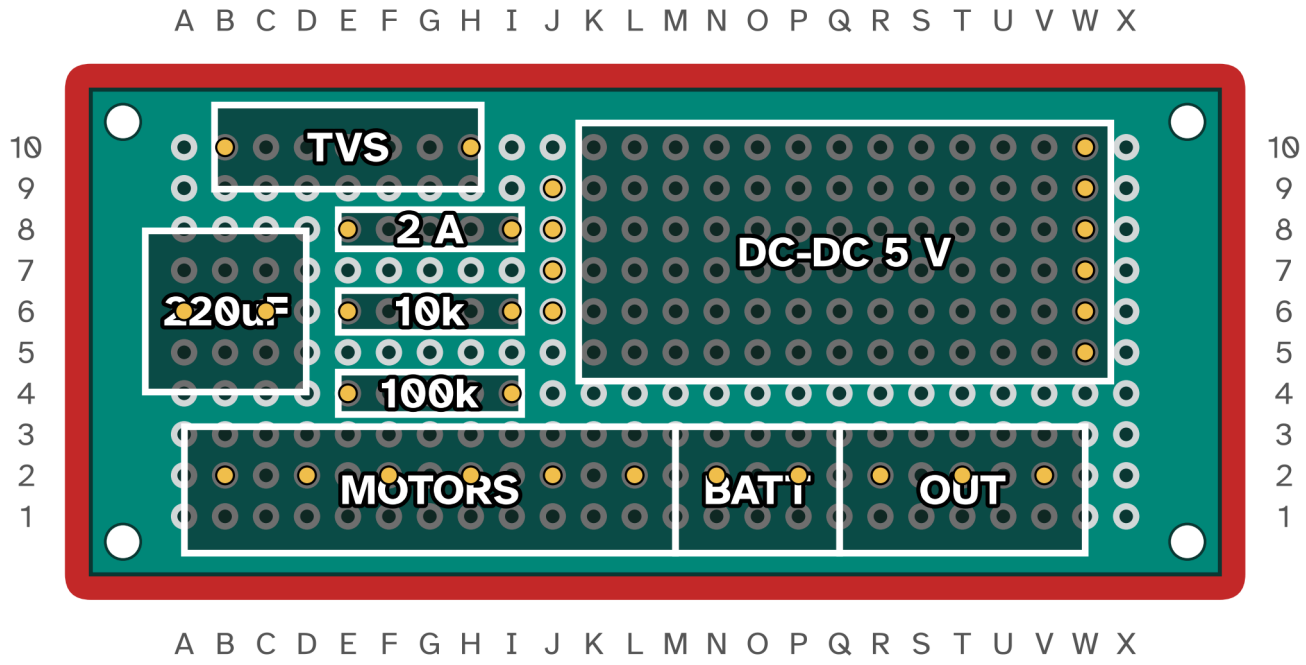
board z · the parts



- **1x Terminal block, 6 poles (2 x 5.08 mm 3-pole)**
  motors: M1+ M1- M2+ M2- M3+ M3- · holes: B2 … L2
- **1x Terminal block, 2 poles (5.08 mm 2-pole)**
  battery: + and - · holes: N2 +, P2 -
- **1x Terminal block, 3 poles (5.08 mm 3-pole)**
  out to y and x: SENSE, GND, +5V · holes: R2, T2, V2
- **1x DC-DC 22.2 V -> 5 V module**
  turned over (marking up), its pins in a female header in row W; a wire from each of its 4 screw poles straight down · holes: GND W10-W8, VOUT W7-W5; poles GND J9, J8, VIN J7, J6
- **1x Pin header, female, 1 x 6, 8.5 mm**
  holds the DC-DC · holes: W10 … W5
- **1x TVS diode 1.5KE33CA - bidirectional: either way round**
  protects +22V from spikes · holes: B10 - H10
- **1x Capacitor 220 uF 50 V - polarized**
  + leg (long) in C6, the stripe (-) side in A6 · holes: C6 +, A6 -
- **1x Fuse 2 A, axial (green, 8 mm), soldered**
  protects the DC-DC · holes: E8 - I8
- **1x Resistor 100 kOhm, 1/4 W**
  voltage divider for SENSE: + side · holes: E4 - I4
- **1x Resistor 10 kOhm, 1/4 W**
  voltage divider for SENSE: GND side · holes: E6 - I6

board z · where the parts go

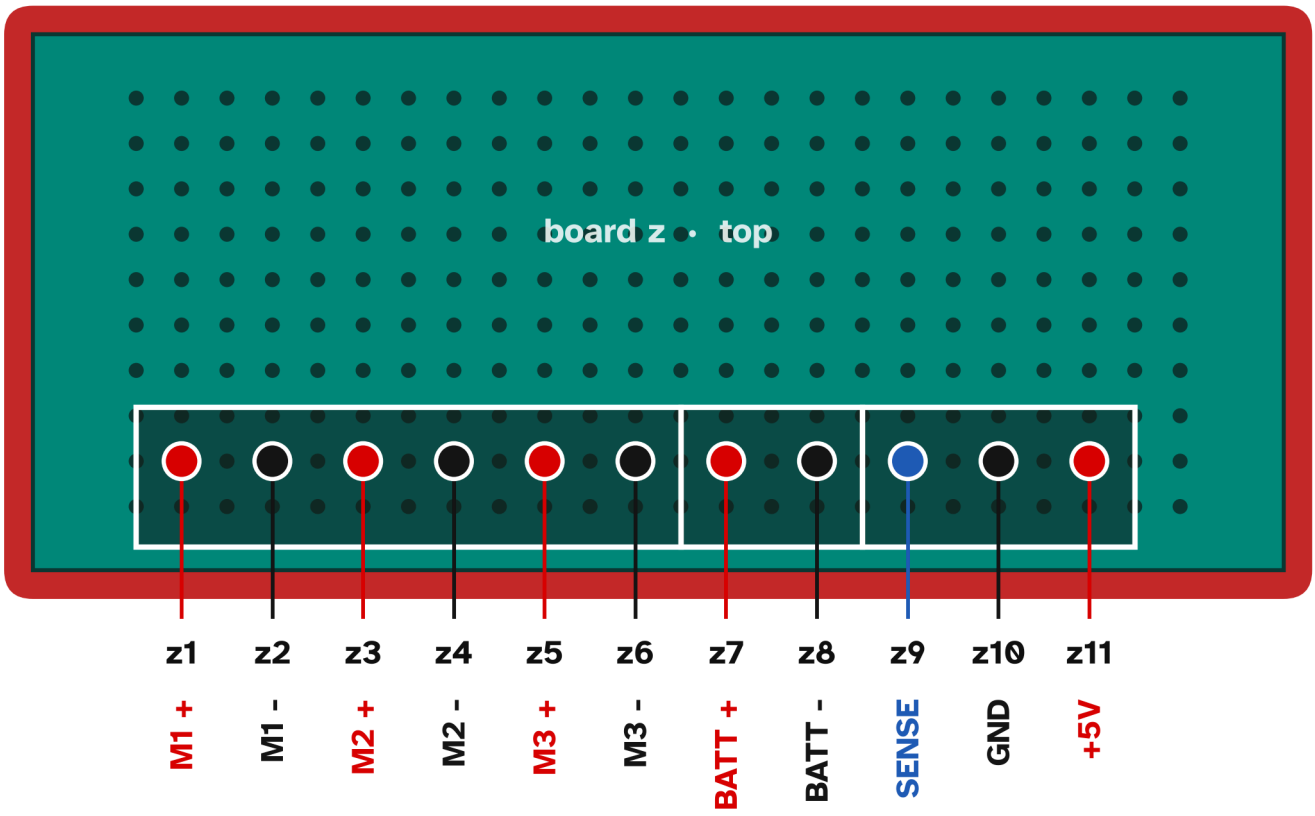
Top side. Yellow dots = the pins.



- Capacitor (polarized): the long leg (+) in C6, the stripe (-) side in A6. Turned round it can burst.
- Terminal blocks: openings face the edge.
- DC-DC: turned over, marking up. Its pins go into the female header in row W.
- Tighten the 4 screw poles BEFORE it goes in: they face the board.
- Screw poles: one wire each, straight down - GND into J9 and J8, VIN into J7 and J6.

board z · the terminals

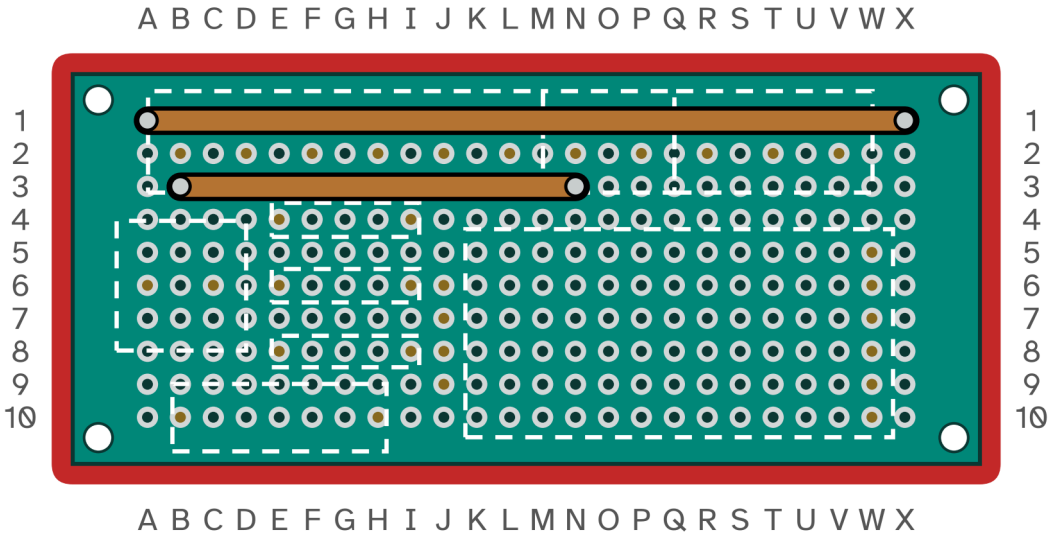
Each pole has a name: the board and a number. The Wiring guide uses the same names.



pole	hole	job	the wire goes to
z1	B2	M1 +	driver X
z2	D2	M1 -	driver X
z3	F2	M2 +	driver Z
z4	H2	M2 -	driver Z
z5	J2	M3 +	driver Y
z6	L2	M3 -	driver Y
z7	N2	BATT +	fuse
z8	P2	BATT -	battery connector -
z9	R2	SENSE	y3
z10	T2	GND	y2, x2
z11	V2	+5V	y1, x1

board z · Bare copper bus bars

Bottom side. Faded = already done.

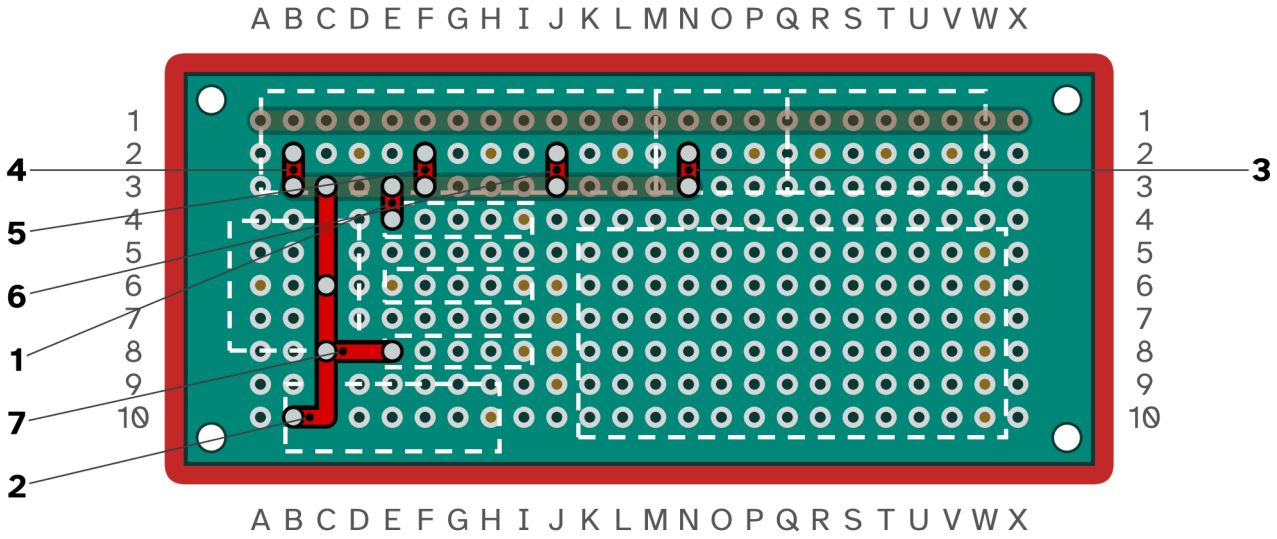


Bare 1.5 mm2 copper, filled with solder, soldered to every hole it passes.

wire	color	from	to	length
	copper	B3 to N3		30 mm
	copper	A1 to X1		58 mm

board z · Red wires: +22V

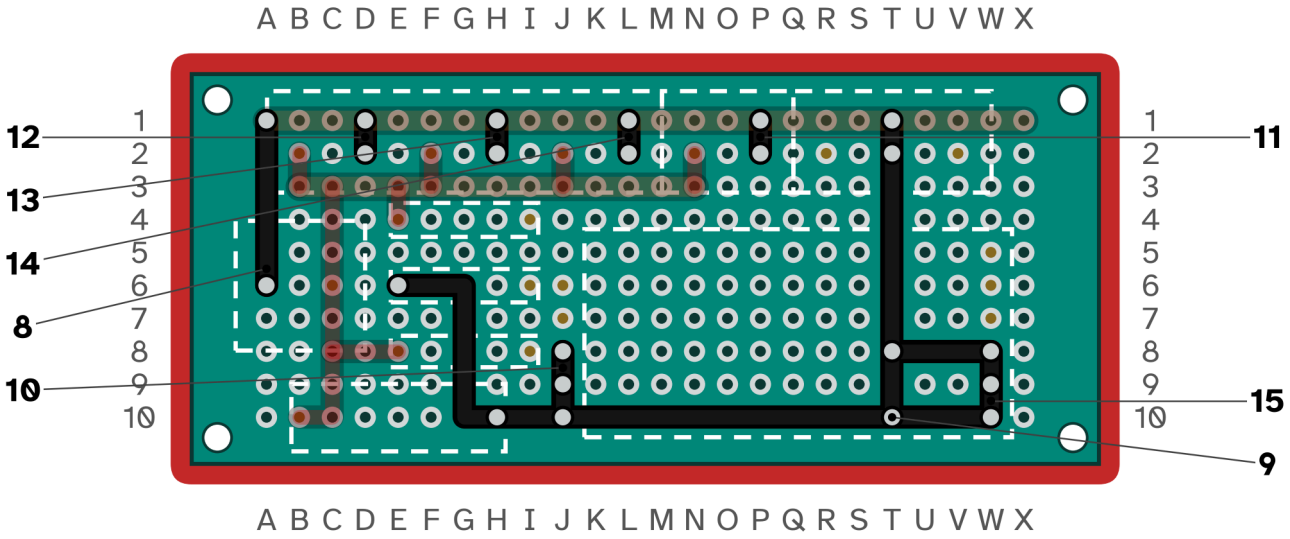
Bottom side. Faded = already done.



wire	color	from	to	length
1	red	E4 100k a	E3 bus bar	3 mm
2	red	B10 TVS A	C3 bus bar	20 mm
3	red	N2 battery +	N3 bus bar	3 mm
4	red	B2 motors M1+	B3 bus bar	3 mm
5	red	F2 motors M2+	F3 bus bar	3 mm
6	red	J2 motors M3+	J3 bus bar	3 mm
7	red	E8 fuse in	C8 on wire 2	5 mm

board z · Black wires: GND

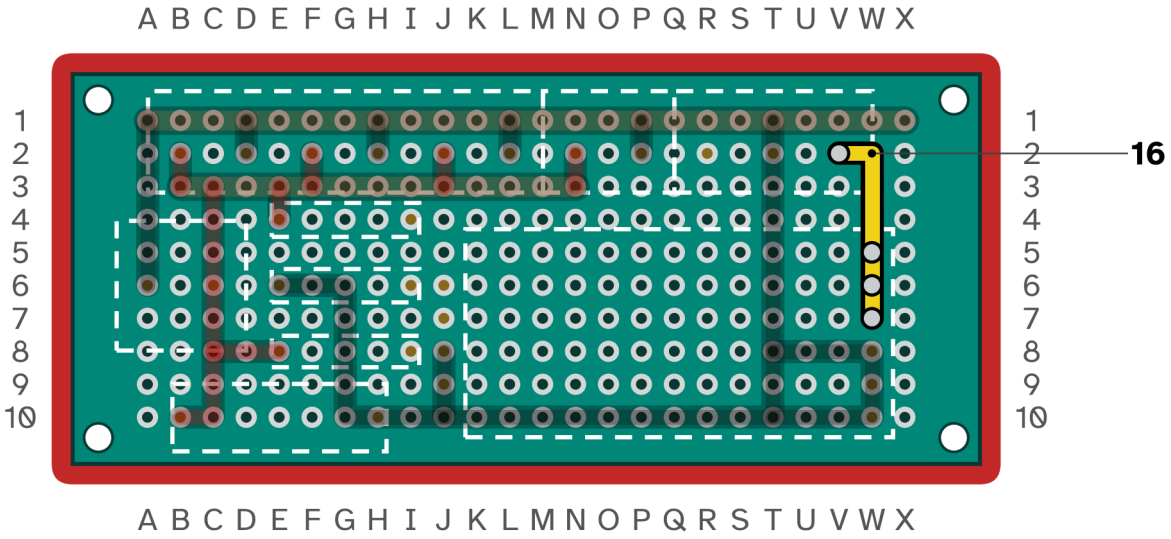
Bottom side. Faded = already done.



wire	color	from	to	length
8	black	A6 cap. -	A1 bus bar	13 mm
9	black	E6 10k b	T1 bus bar	71 mm
10	black	J8 DC-DC pole GND2	J10 on wire 9	5 mm
11	black	P2 battery -	P1 bus bar	3 mm
12	black	D2 motors M1-	D1 bus bar	3 mm
13	black	H2 motors M2-	H1 bus bar	3 mm
14	black	L2 motors M3-	L1 bus bar	3 mm
15	black	T10 on wire 9	T8 on wire 9	20 mm

board z · Yellow wires: +5V

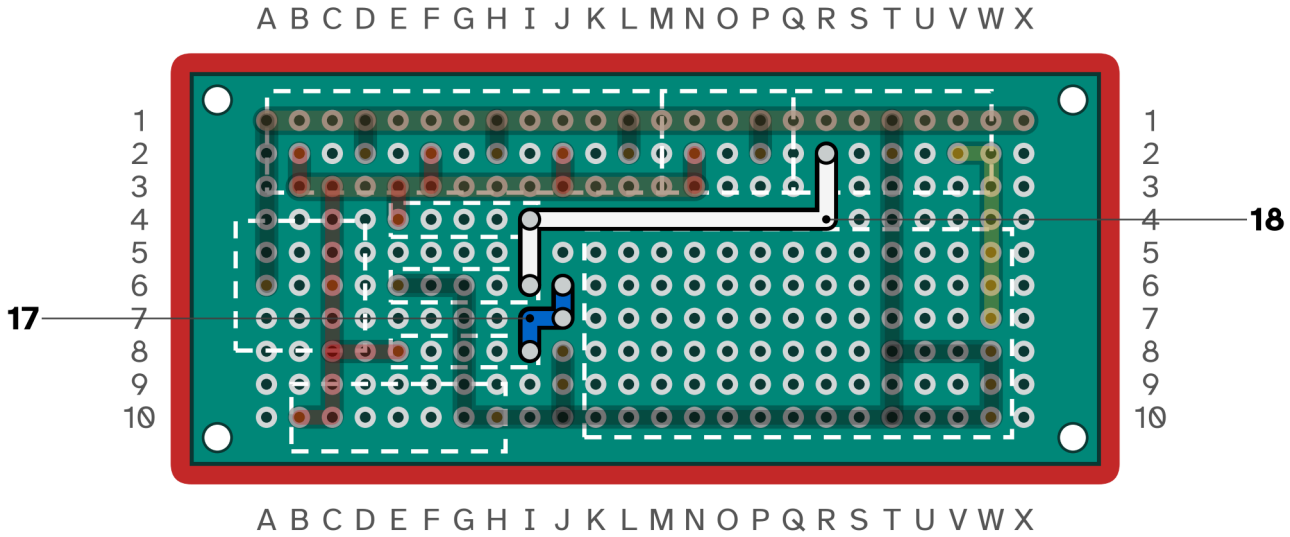
Bottom side. Faded = already done.



wire	color	from	to	length
16	yellow	V2 output +	W7 DC-DC VO1	15 mm

board z · Signal wires

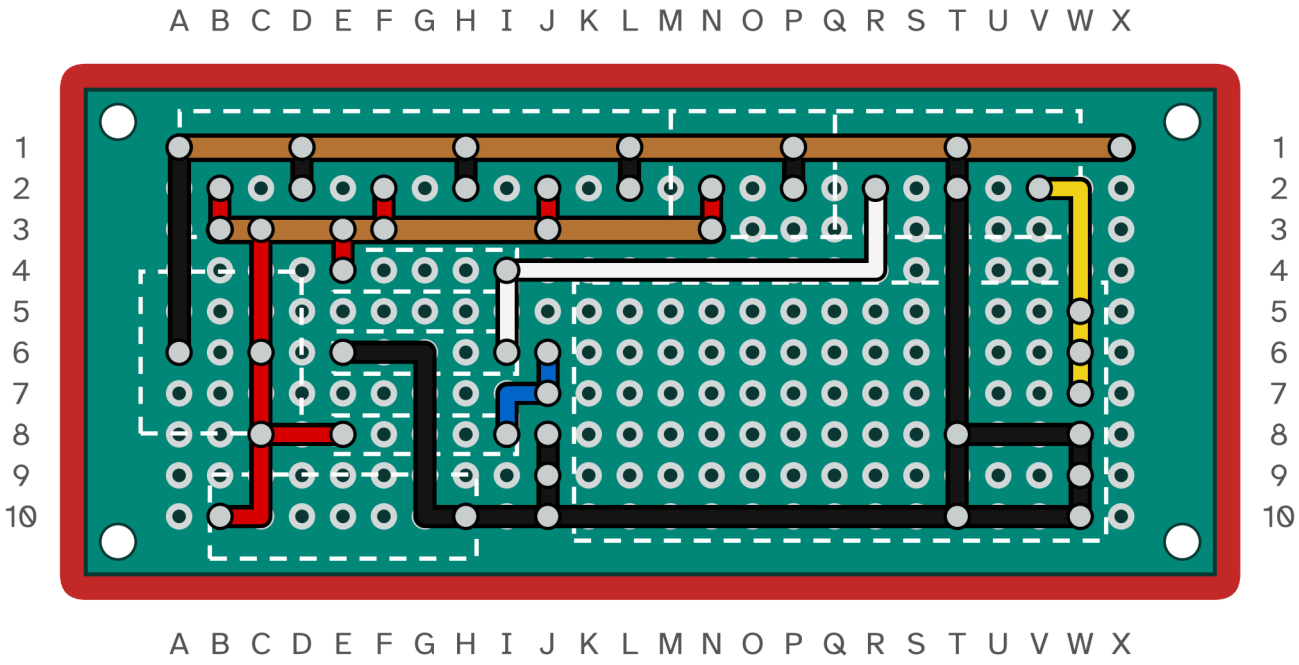
Bottom side. Faded = already done.

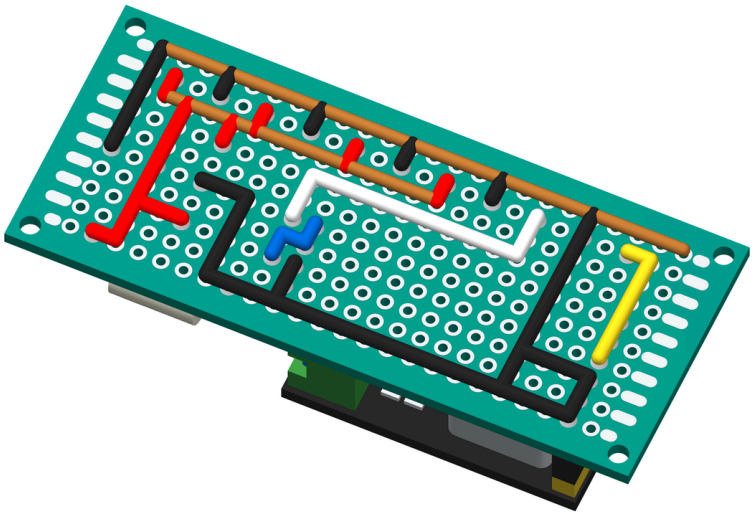


wire	color	from	to	length
17	blue	J6 DC-DC pole VIN2	I8 fuse out	8 mm
18	white	I6 10k a	R2 output S	33 mm

board z · done: like this

All the wires, bottom side.





board z · check: must beep

Multimeter on the beep setting. Board NOT connected.

- **+22V**: “N2 - B2 - F2 - J2 - B10 - C6 - E4 - E8”
- **GND**: “P2 - D2 - H2 - L2 - H10 - A6 - E6 - T2 - W10 - W9 - W8 - J9 - J8”
- **+22V after the fuse**: “I8 - J7 - J6”
- **+5V**: “W7 - W6 - W5 - V2”
- **SENSE**: “I4 - I6 - R2”

Put one probe on the first hole, then touch each of the others: every one must beep.


beep setting


not connected


must beep

board z · check: must NOT beep

Multimeter on the beep setting. Board NOT connected.

- +22V (N2) and GND (P2)
- +22V (N2) and +5V (W7)
- GND (P2) and +5V (W7)

A beep here is a short circuit: find it before you connect anything.


beep setting

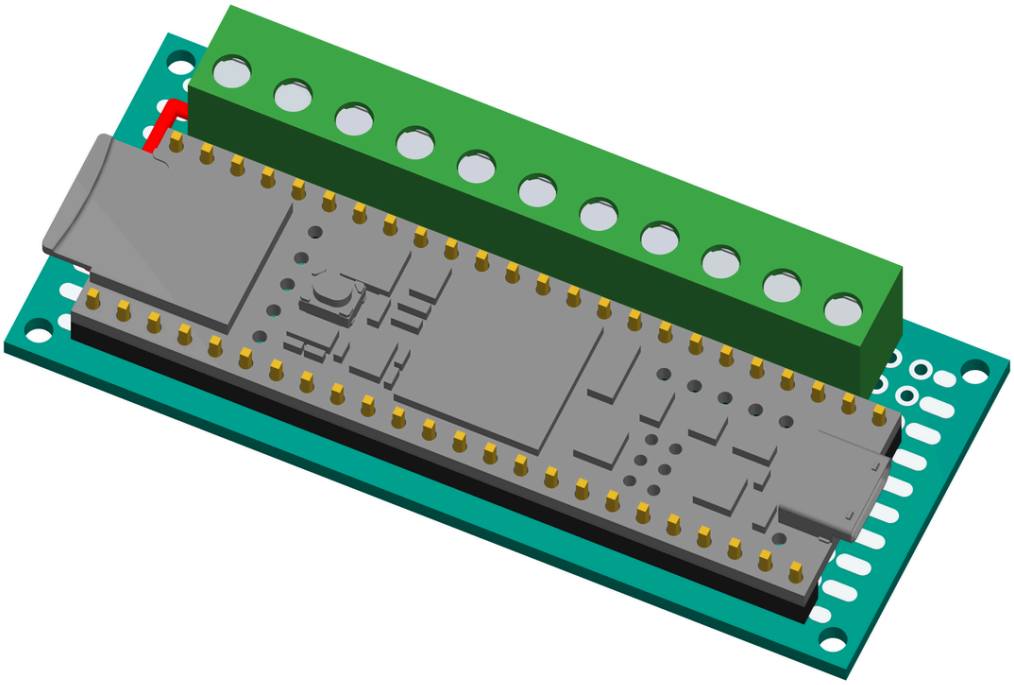

not connected


must NOT beep

board y · MCU

The brain: the MCU board reads the IMU and the battery voltage and decides how the wheels must turn.

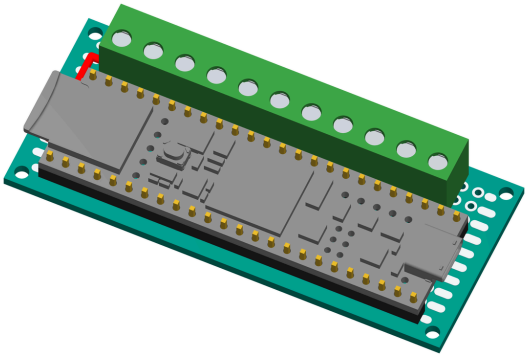
board y



6 parts · 13 wires

Face y of the box. Find the letter Y on the box wall.

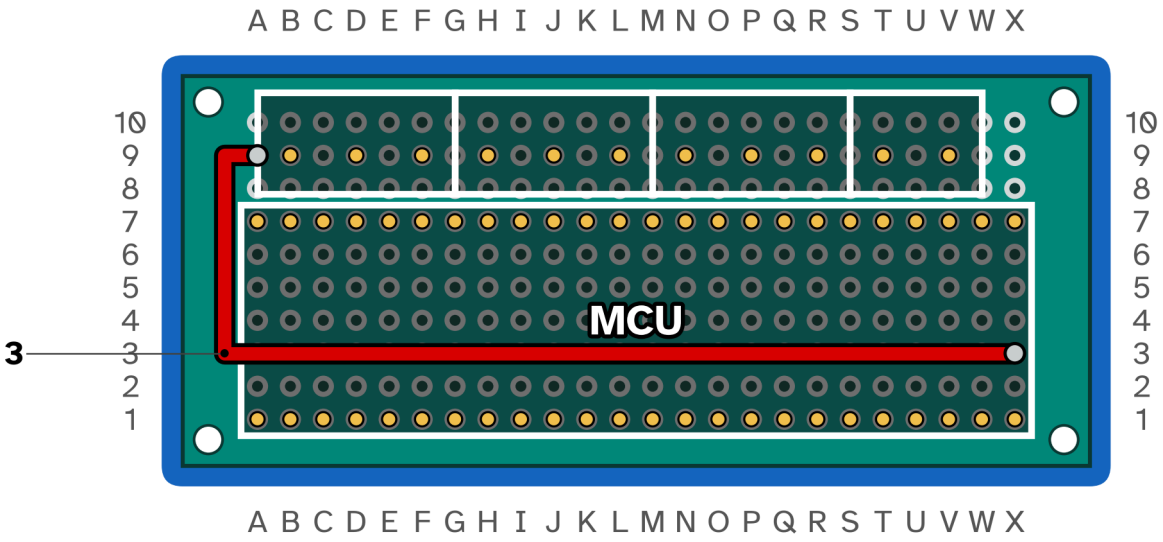
board y · the parts



- **1x MCU board on 2 pin headers (24 pins each)**
  USB at row X · holes: columns 1 and 7, rows A ... X
- **4x Terminal blocks: 3 x 5.08 mm 3-pole + 1 x 5.08 mm 2-pole**
  one line of 11 poles · holes: column 9, rows B ... V
- **1x Red insulated wire, thin: 24 AWG (about 1.2 mm with the insulation)**
  +5V on the TOP side, under the MCU - 89 mm · holes: X3 - A9

board y · where the parts go

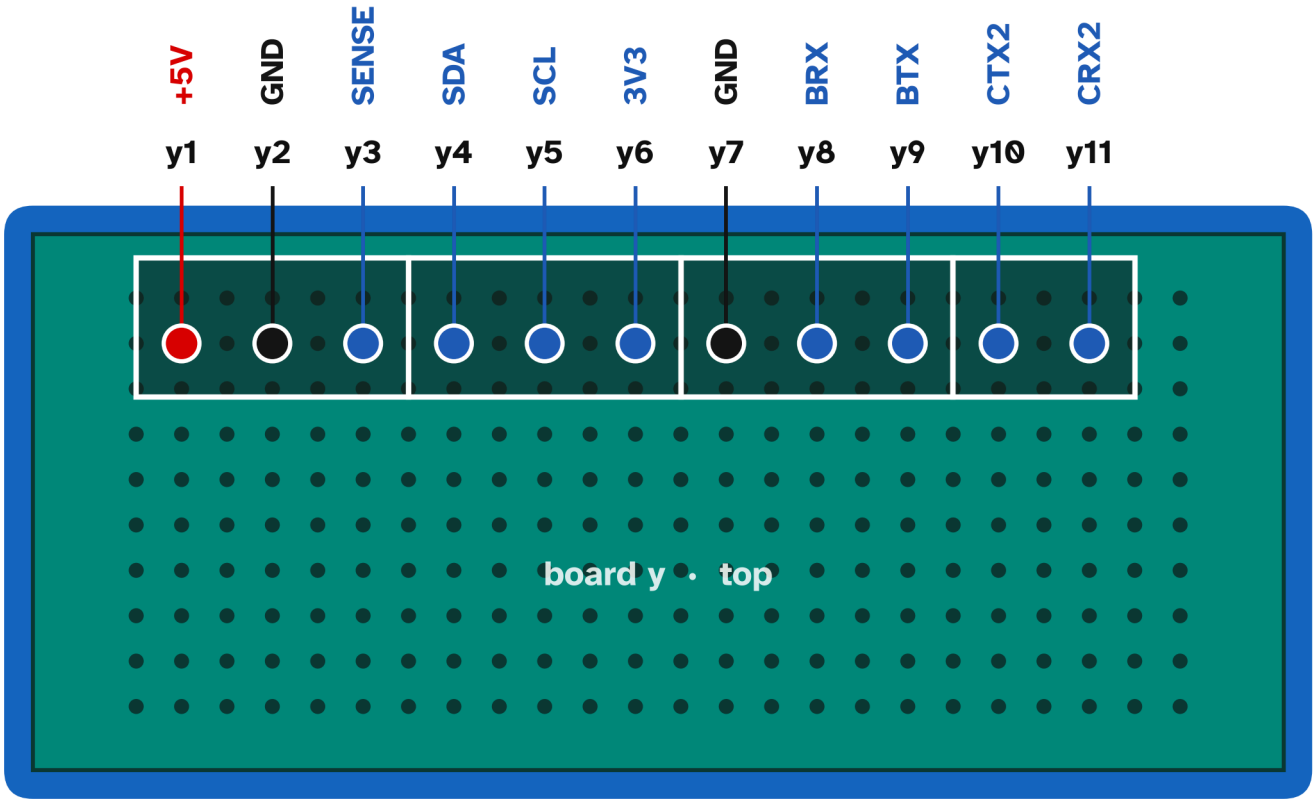
Top side. Yellow dots = the pins.



- MCU: USB at row X, pins in columns 1 and 7.
- Terminal blocks: openings face the edge, column 9.
- Red wire 3 (+5V): the only insulated wire. Lay it BEFORE the MCU: from X3 along row 3, flat on the board (the MCU will sit over it), out past column A, down to A9. Its ends go down through the holes; you solder them from below, on wires 4 and 5.

board y · the terminals

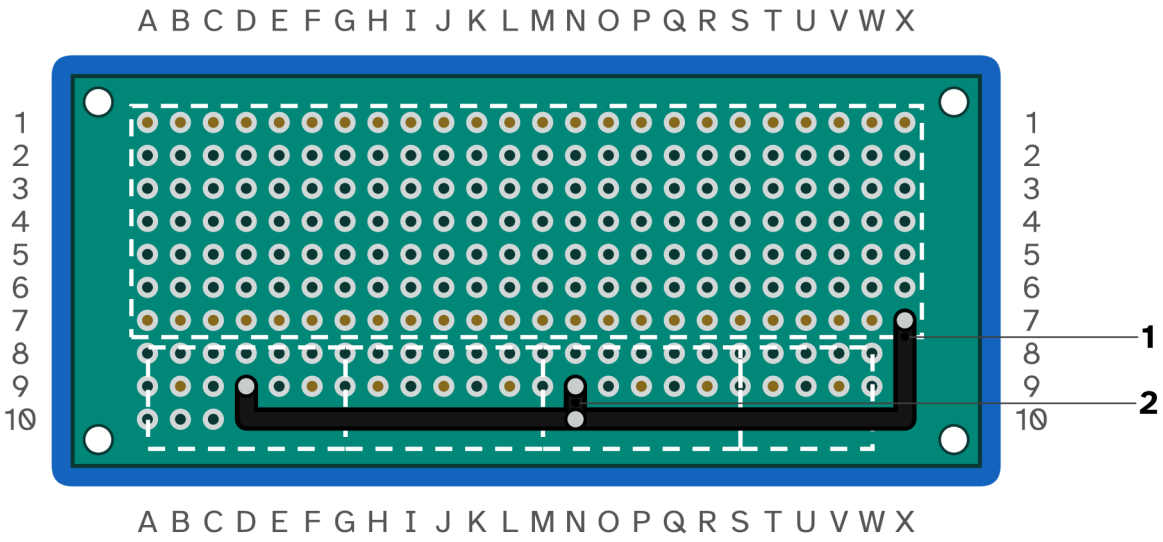
Each pole has a name: the board and a number. The Wiring guide uses the same names.



pole	hole	job	the wire goes to
y1	B9	+5V	z11
y2	D9	GND	z10
y3	F9	SENSE	z9
y4	H9	SDA	IMU
y5	J9	SCL	IMU
y6	L9	3V3	IMU
y7	N9	GND	IMU
y8	P9	BRX	x3
y9	R9	BTX	x4
y10	T9	CTX2	x6
y11	V9	CRX2	x7

board y · Black wires: GND

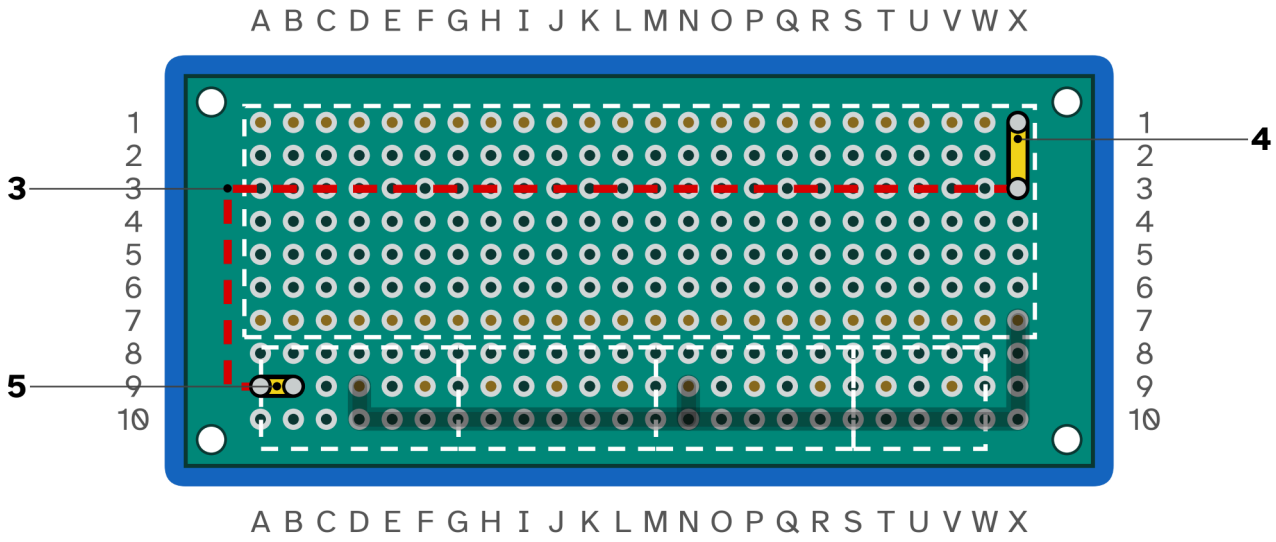
Bottom side. Faded = already done.



wire	color	from	to	length
1	black	X7 MCU GND	D9 block GND	61 mm
2	black	N9 block GND_I	N10 on wire 1	3 mm

board y · Yellow wires: +5V

Bottom side. Faded = already done.

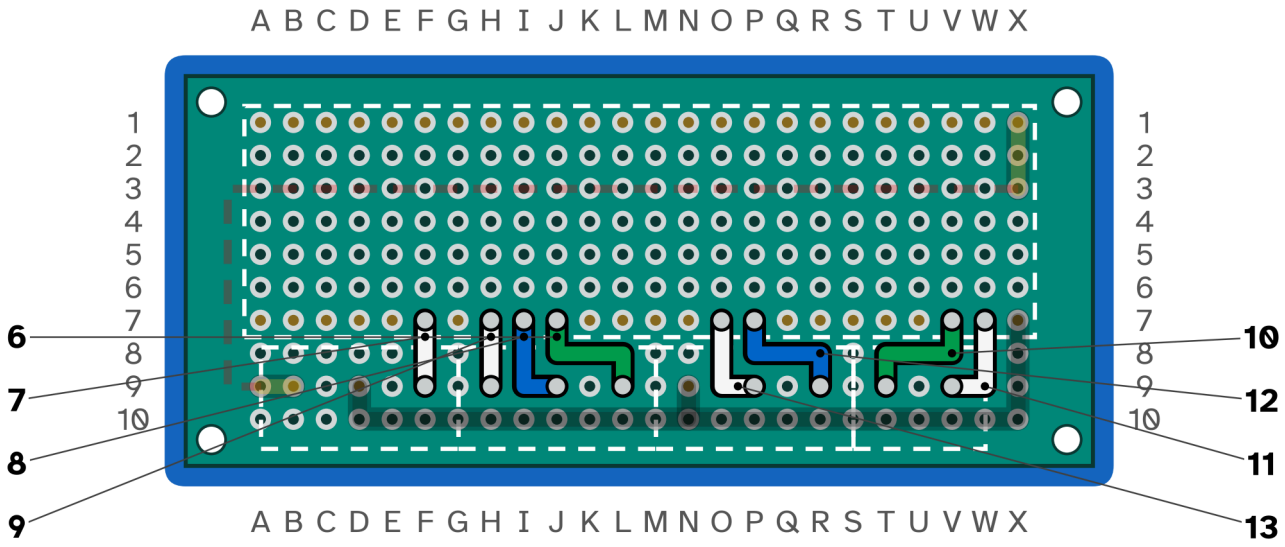


wire	color	from	to	length
3	red	X3 top side	A9 top side	89 mm
4	yellow	X1 MCU VIN	X3 on wire 3	5 mm
5	yellow	A9 on wire 3	B9 block +5V	3 mm

- Wire 3 is the red insulated wire on the TOP side (dashed here): laid before the MCU. Here you only solder its two ends, X3 and A9, together with the bare wires that meet there.

board y · Signal wires

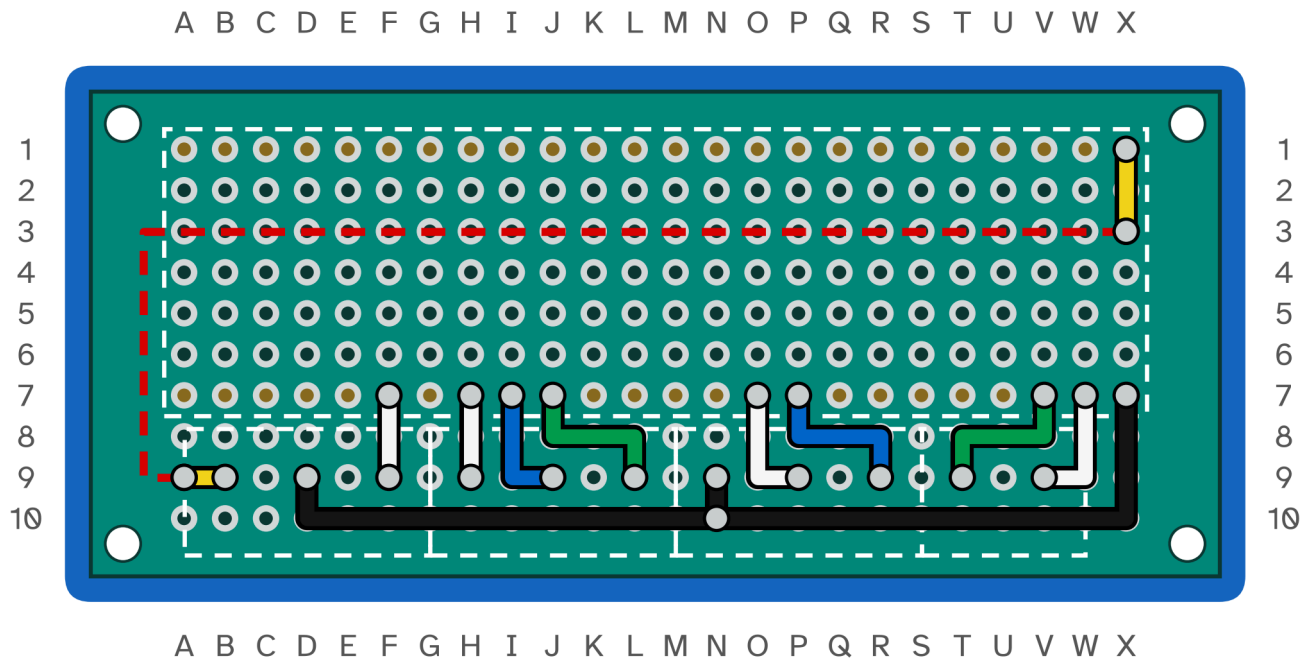
Bottom side. Faded = already done.

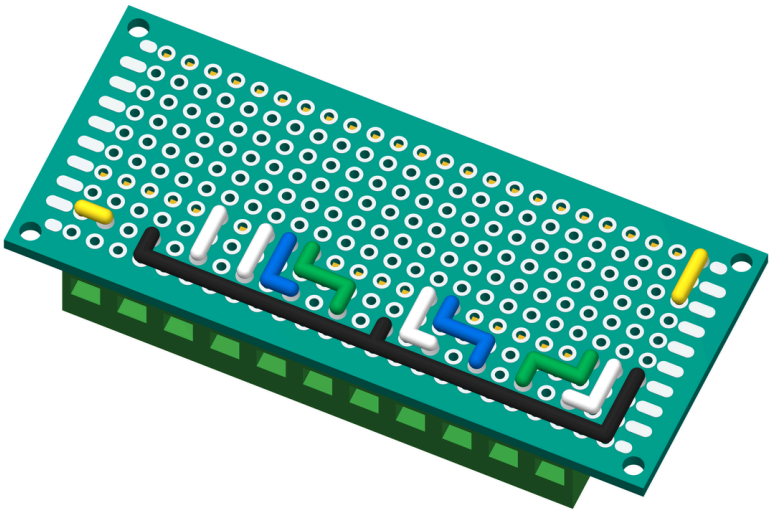


wire	color	from	to	length
6	green	J7 MCU 3V3b	L9 block 3V3	10 mm
7	white	F7 MCU 27	F9 block SENSE	5 mm
8	blue	I7 MCU 24	J9 block SCL	8 mm
9	white	H7 MCU 25	H9 block SDA	5 mm
10	green	T9 block CTX2	V7 MCU 1	10 mm
11	white	V9 block CRX2	W7 MCU 0	8 mm
12	blue	P7 MCU 7	R9 block BTX	10 mm
13	white	O7 MCU 8	P9 block BRX	8 mm

board y · done: like this

All the wires, bottom side.





board y • check: must beep

Multimeter on the beep setting. Board NOT connected.

- **+5V**: “B9 - X1”
- **GND**: “D9 - N9 - X7”
- **3V3**: “J7 - L9”
- **SENSE**: “F9 - F7”
- **SCL2**: “J9 - I7”
- **SDA2**: “H9 - H7”
- **CTX2**: “V7 - T9”
- **CRX2**: “W7 - V9”
- **Wi-Fi board TX>RX2**: “R9 - P7”
- **Wi-Fi board RX<TX2**: “P9 - O7”

Put one probe on the first hole, then touch each of the others: every one must beep.


beep setting


not connected


must beep

board y · check: must NOT beep

Multimeter on the beep setting. Board NOT connected.

- +5V (B9) and GND (D9)

A beep here is a short circuit: find it before you connect anything.


beep setting

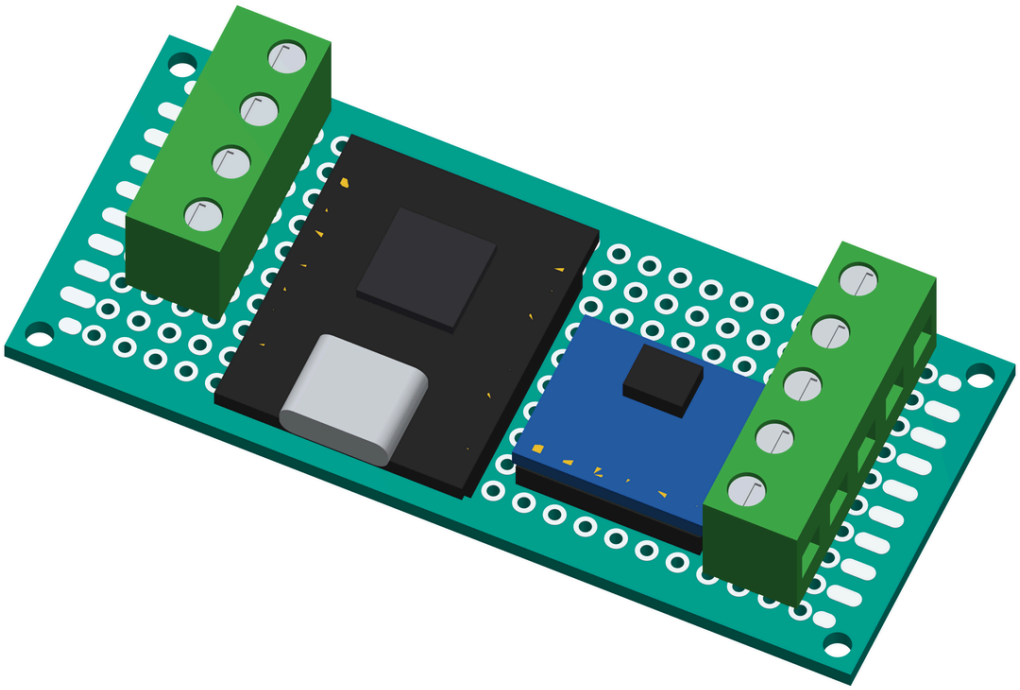

not connected


must NOT beep

board x · WIRELESS + CAN

Talks to your laptop by radio (Wi-Fi board) and to the 3 motor controllers over the CAN bus (CAN board).

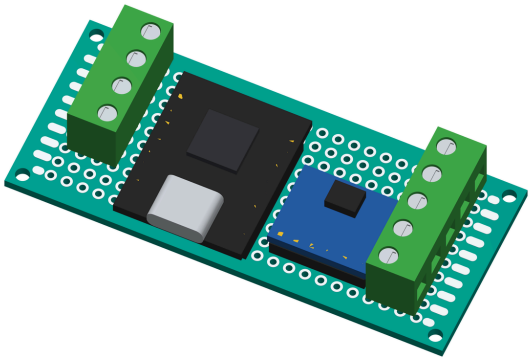
board x



6 parts · 9 wires

Face x of the box. Find the letter X on the box wall.

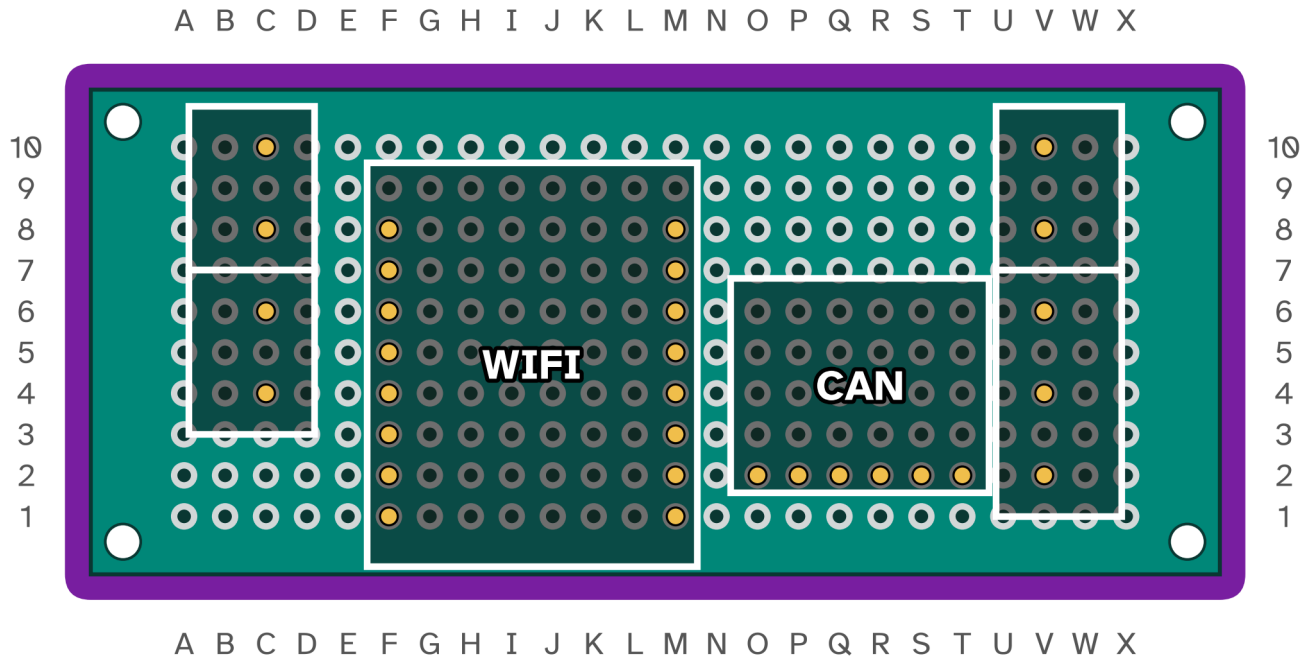
board x · the parts



- **2x Terminal blocks: 2 x 5.08 mm 2-pole**
  +5V, GND, B-TX, B-RX · holes: C10 … C6
- **1x Wi-Fi board, turned (on 2 x 8 pins)**
  USB toward row 1; solder by the names on its board · holes: F1 … F8 and M1 … M8
- **1x CAN transceiver board (on 6 pins)**
  3V3 GND CTX CRX CANH CANL · holes: O2 … T2
- **2x Terminal blocks: 5.08 mm 3-pole + 5.08 mm 2-pole**
  CANL, CANH, CRX2, CTX2, GND · holes: V2 … V10

board x · where the parts go

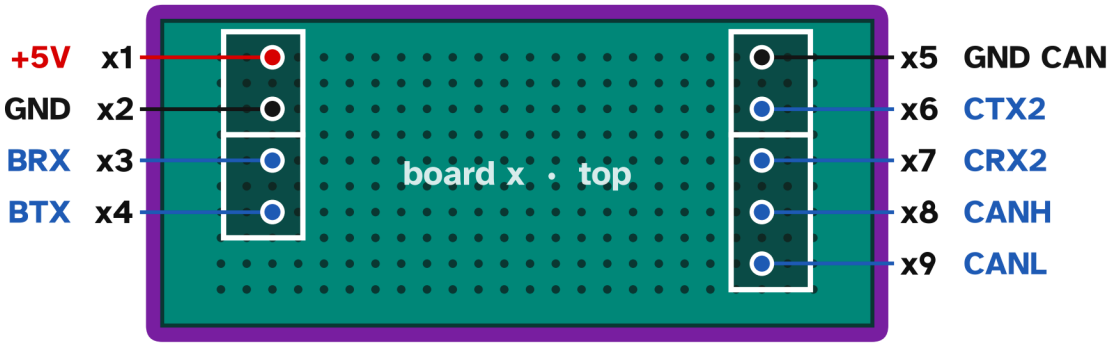
Top side. Yellow dots = the pins.



- Wi-Fi board: turned, its USB toward row 1. Its BAT pin in F1, GND2 in M1 - check the names printed on it before you solder.
- CAN board: 3V3 pin in O2 … CANL in T2, as printed on it.
- Terminal blocks: one row in from the edge, openings out.

board x · the terminals

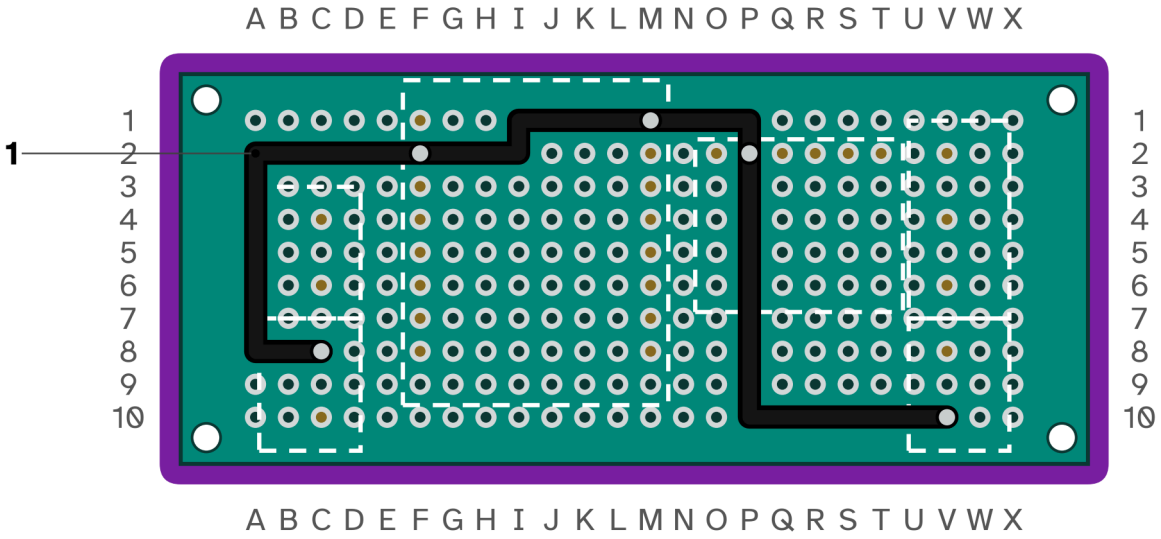
Each pole has a name: the board and a number. The Wiring guide uses the same names.



pole	hole	job	the wire goes to
x1	C10	+5V	z11
x2	C8	GND	z10
x3	C6	BRX	y8
x4	C4	BTX	y9
x5	V10	GND CAN	CAN 1 red
x6	V8	CTX2	y10
x7	V6	CRX2	y11
x8	V4	CANH	CAN 1 yellow
x9	V2	CANL	CAN 1 black

board x · Black wires: GND

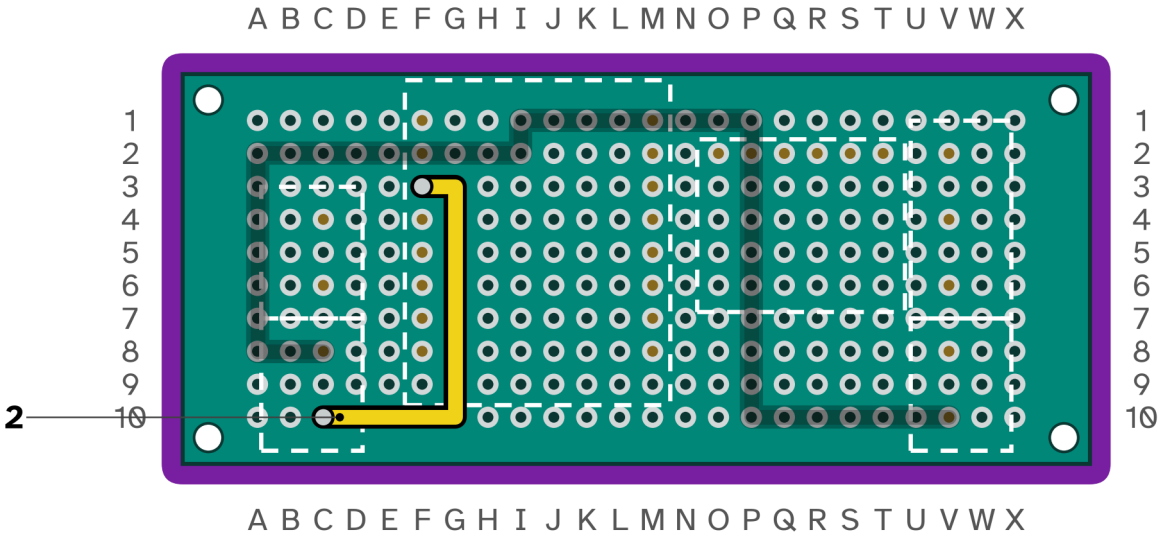
Bottom side. Faded = already done.



wire	color	from	to	length
1	black	V10 block GNDC	C8 block GND	99 mm

board x · Yellow wires: +5V

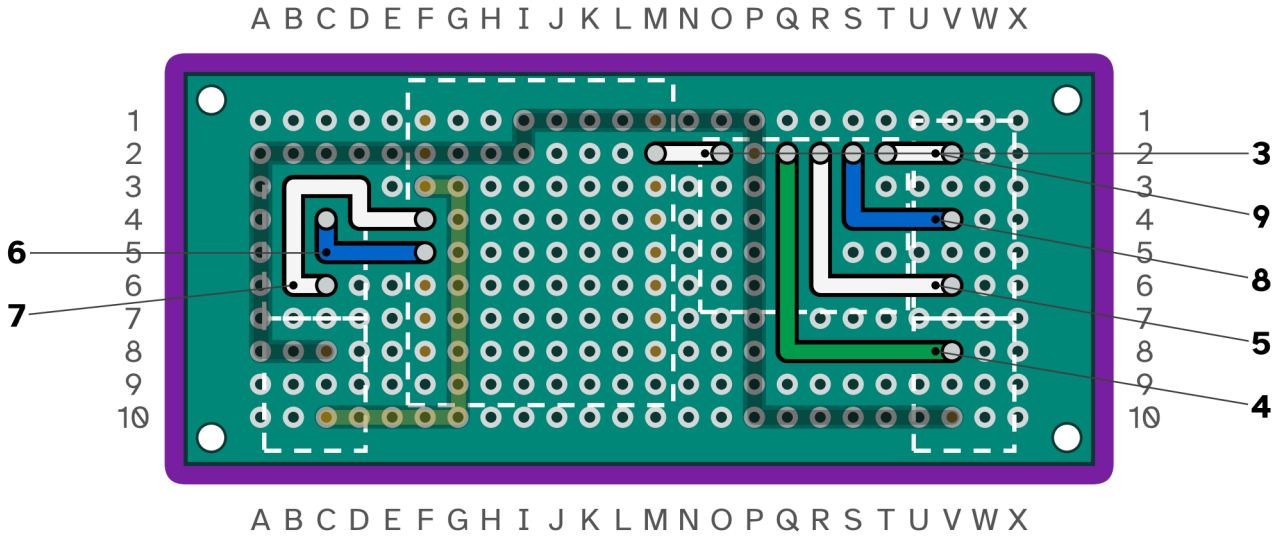
Bottom side. Faded = already done.



wire	color	from	to	length
2	yellow	C10 block +5V	F3 Wi-Fi board VIN	30 mm

board x · Signal wires

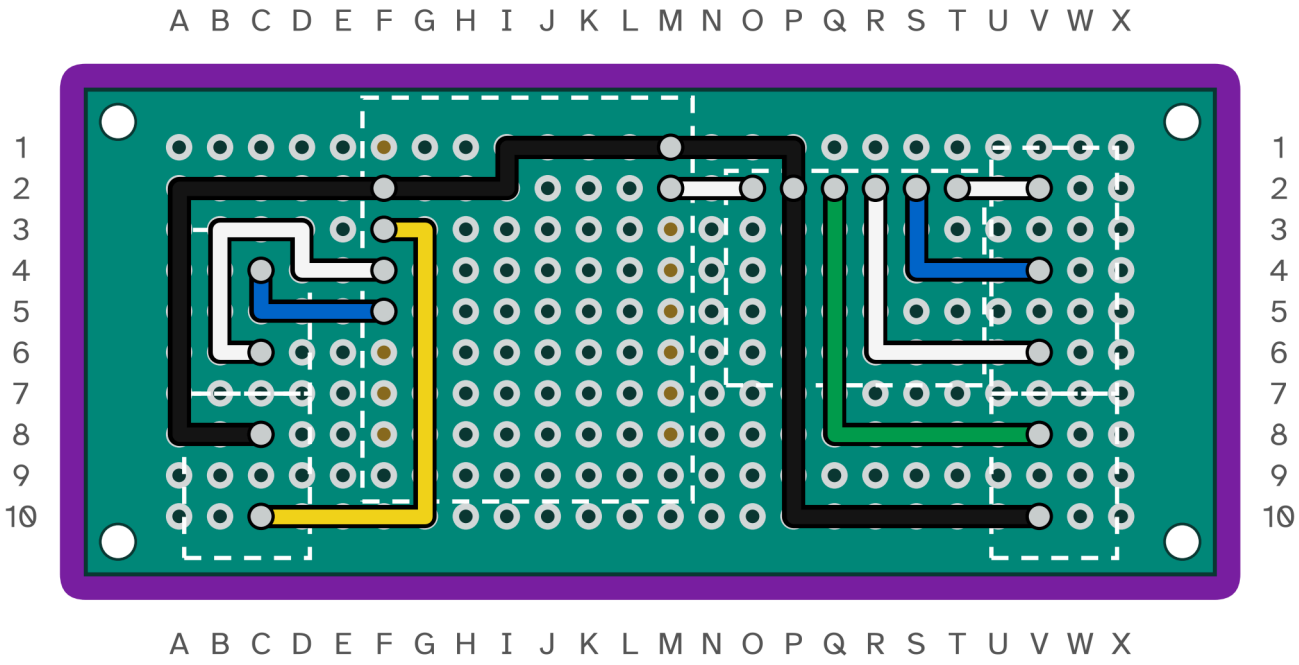
Bottom side. Faded = already done.

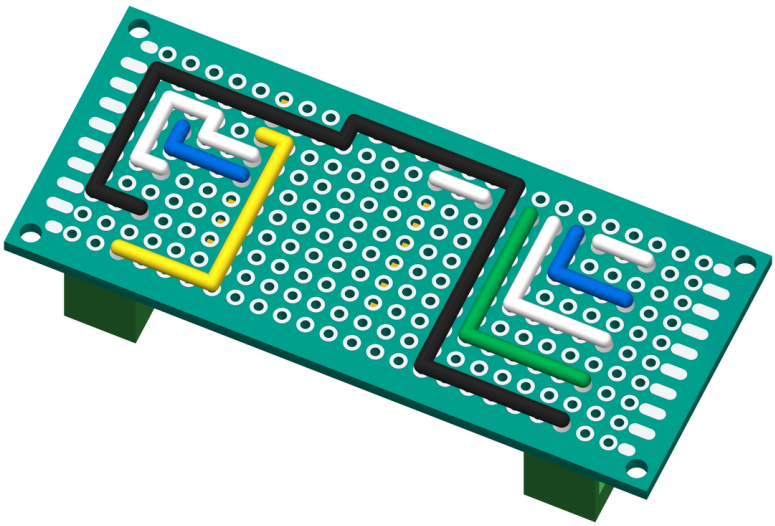


wire	color	from	to	length
3	white	O2 CAN 3V3	M2 Wi-Fi board 3V3	5 mm
4	green	Q2 CAN CTX	V8 block CTX2	28 mm
5	white	R2 CAN CRX	V6 block CRX2	20 mm
6	blue	F5 Wi-Fi board TX16	C4 block BTX	10 mm
7	white	C6 block BRX	F4 Wi-Fi board RX17	23 mm
8	blue	S2 CAN CANH	V4 block CANH	13 mm
9	white	V2 block CANL	T2 CAN CANL	5 mm

board x · done: like this

All the wires, bottom side.





board x · check: must beep

Multimeter on the beep setting. Board NOT connected.

- **+5V**: “C10 - F3”
- **GND**: “C8 - F2 - M1 - P2 - V10”
- **3V3 Wi-Fi board**: “M2 - O2”
- **CTX2**: “Q2 - V8”
- **CRX2**: “R2 - V6”
- **Wi-Fi board TX16**: “F5 - C4”
- **Wi-Fi board RX17**: “F4 - C6”
- **CANH**: “S2 - V4”
- **CANL**: “T2 - V2”

Put one probe on the first hole, then touch each of the others: every one must beep.


beep setting


not connected


must beep

board x · check: must NOT beep

Multimeter on the beep setting. Board NOT connected.

- +5V (C10) and GND (C8)

A beep here is a short circuit: find it before you connect anything.


beep setting


not connected


must NOT beep

Good to know



Not sure?

Look at the drawing again. Count the holes from the letter and the number.



Check twice

The multimeter finds what the eye misses: beep where it must, silence where it must not.